



Do key audit matter disclosures about M&A transactions predict future performance?

Jessica A. Nysten¹ · Daniel D. Wangerin¹ · Karla M. Zehms¹

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Abstract

This study examines whether auditor KAM disclosures about M&A transactions predict future performance outcomes. We find that auditors make more M&A KAM disclosures when M&A transactions are larger, involve targets with greater information asymmetry, and are subject to greater overpayment risk. When auditors disclose M&A KAMs in the acquisition year, we find an increased risk of larger goodwill impairments and that the association between acquired goodwill and changes in companies' subsequent operating performance are significantly lower compared to when auditors do not issue an M&A KAM. We also find that the likelihood of forced CEO turnover is incrementally higher when subsequent operating performance is poor and auditors issue an M&A KAM compared to when they do not. Collectively, our findings suggest that these auditor disclosures reflect uncertainty that predicts weak post-acquisition performance and forced CEO turnover, incremental to other indicators of failed acquisition strategies.

Keywords CEO turnover · Goodwill · Key audit matters · Mergers and acquisitions · Post-acquisition performance · Uncertainty

JEL Classification D82 · G34 · M41

✉ Daniel D. Wangerin
daniel.wangerin@wisc.edu

Jessica A. Nysten
jnylen@wisc.edu

Karla M. Zehms
karla.zehms@wisc.edu

¹ Wisconsin School of Business, University of Wisconsin-Madison, Madison, WI, USA

1 Introduction

Mergers and acquisitions (M&A) represent the largest investments that many companies make—global transaction volume averages around \$4 trillion per year. Successful acquisitions contribute to growth and shareholder returns, while failed acquisitions destroy shareholder value. Despite this importance, there is substantial uncertainty about whether any particular transaction will succeed. Investors and analysts have difficulty predicting and evaluating post-acquisition performance, motivating recent calls to improve disclosures about M&A transactions (IASB 2020, 2024). CEOs face strong pressure to deliver post-acquisition profits, creating incentives for opportunistic reporting (Bens et al. 2012). When M&A transactions fail to deliver expected results, investors and analysts often claim that companies delay recognizing goodwill impairments until long after managers become aware of valuation declines.¹ Consequently, managers' financial reporting disclosures may not be a reliable source to distinguish which M&A transactions will prove to be a success versus a failure.

An alternative source of reliable information derives from audit reports. Regulators require that auditors provide key audit matter (KAM) disclosures, which reveal the most significant matters in the audit that involve judgment and high estimation uncertainty. We examine acquisition characteristics associated with auditors' M&A KAM disclosures that communicate information about the uncertainty implicit in M&A transactions. We then investigate whether auditors' M&A KAM disclosures predict the future success or failure of managers' acquisition strategies. We focus on KAMs that identify M&A-related risks in the audited acquisition-year financial statements and describe the procedures auditors use to evaluate and respond to those risks (e.g., evaluating the effectiveness of relevant internal controls, reviewing legal agreements, and testing assumptions in valuation models). Under the historical pass/fail auditor reporting regime, audit opinions were an inadequate means of expressing auditors' uncertainty about valuation judgments inherent in the accounting for M&A transactions. KAM disclosures now provide auditors the opportunity to communicate their judgments about uncertainty via their audit reports (FRC 2013, 2016a, 2016b; Rousseau and Zehms 2024).

Despite auditors' privileged position with respect to private information about M&A transactions (Sheridan 2015), it is unclear whether M&A KAM disclosures reflect useful information predictive of the future success or failure of managers' acquisition strategies. Auditing for M&A transactions (i.e., business combinations) has consistently been among the most frequently occurring audit deficiencies that the PCAOB has identified (e.g., see the 2022 and 2023 summaries of inspection observations (PCAOB 2023, 2024)). In light of auditors' prolonged difficulty evaluating

¹ In its 2020 comment letter to the FASB on goodwill and intangible assets, the CFA Institute explained its view that goodwill impairments can provide valuable information about unsuccessful acquisitions but that current U.S. GAAP provides inadequate disclosure, masks poor performance, and lacks timeliness (CFA Institute 2020). These views are consistent with prior research that shows managers use financial reporting discretion to avoid or delay recognizing goodwill impairments (Ramanna and Watts 2012; Li and Sloan 2017; Ayres et al. 2019).

these particularly complex transactions, it is possible that auditors' disclosures will fail to effectively convey uncertainty about managers' acquisition strategies.² More generally, studies on the capital market consequences of KAM reporting suggest auditors' disclosures lack information content and only confirm information that investors have already impounded in equity prices (Bédard et al. 2019; Lennox et al. 2023; Burke et al. 2023; Liao et al. 2024). These findings are consistent with the view that KAMs fail to provide investors with meaningful information and that auditors make boilerplate disclosures in response to litigation risk rather than investors' informational needs.

To conduct our study, we examine a sample of acquisitions that companies listed on the London Stock Exchange completed between 2013 and 2020. We then conduct a series of analyses, following each company in our sample beginning in the acquisition year, and ending three years later. Over this period, we examine (1) whether the auditor issues an M&A-related KAM in its audit report for the acquisition year; (2) the extent to which the company realizes (or fails to realize) expected benefits from management's acquisition strategy, as reflected in goodwill impairments and post-acquisition performance in the subsequent three years; and (3) whether the CEO undergoes forced turnover in the subsequent three years.

In our first series of tests, we examine M&A KAMs and the acquisition characteristics that auditors consider in their disclosure decisions. Using data from the Audit Analytics Europe database, we identify M&A KAMs in which the auditor makes a KAM disclosure related to accounting for business combinations in the acquisition completion year. Economic uncertainty about the future success or failure of an M&A transaction arises from such sources as the relative size of the transaction, information asymmetries between the target and acquirer, and valuation uncertainty that contributes to a higher risk of overpayment. Accounting for the transaction in the context of these uncertainties makes auditing around it judgment laden, due to resulting levels of estimation and subjectivity. Consequently, we expect that when M&A transactions possess more uncertainty, the audit report is more likely to include an M&A KAM.

Indeed, we predict and find that auditors are more likely to issue an M&A KAM when acquirers complete transactions that have a material impact on the financial statements compared to when transactions have an immaterial impact. Further, in contrast to criticisms that KAMs lack information content, we also predict and find that auditors' decision to disclose an M&A KAM varies predictably across deals that are more likely to entail greater information asymmetry and overpayment risk, such as in acquisitions involving foreign targets and those where the acquirer

² The Financial Reporting Council (FRC) is the audit standards setter and regulator in the United Kingdom. Like the PCAOB, the FRC writes auditing standards as well as inspects and supervises audit firms. The FRC provides a yearly report detailing overall themes as well as reports on individual audit firms based on the size of their share of the public interest entity market (Tiers 1, 2, and 3). Tier 1 audit firms undergo the most scrutiny. We reviewed the themes the FRC articulated for Tier 1 firms during the period 2022–2024 and found that audit quality detriments for business combinations are less common than in the United States. Instead, revenue and impairments were either the first or second most common (FRC 2022, 2023, 2024).

recognizes more goodwill. In these deals, information asymmetry and adverse selection problems are more-pronounced concerns, and there is greater valuation uncertainty about potential synergies and the expected cash flows from the acquisition implied by the purchase price. As an example of the economic magnitude of these results, our results reveal a 5 percent increase in the likelihood that the audit report contains an M&A KAM for each one standard deviation increase in goodwill. Collectively, these results imply that M&A KAM disclosures reflect auditors' evaluations of the uncertainty underlying their clients' accounting estimates of acquired identifiable net assets and goodwill. In the years following the acquisition, auditors evaluate goodwill for potential impairment and consider the uncertainty of future performance related to the acquired entity.

The second series of tests shifts to the periods following the transaction and examines whether M&A KAM disclosures predict the subsequent success or failure of managers' acquisition strategies. In these tests, we examine (1) future goodwill impairments and (2) how the goodwill recognized in the acquisition predicts subsequent operating performance (measured as changes in the company's operating cash flows over the three years following the acquisition year). We find that, when the auditor issues an M&A KAM in the acquisition year, the acquirer records larger future goodwill impairments and its auditor is more likely to make future KAM disclosures about goodwill than when the auditor does not issue an M&A KAM. We also find that the positive association between the acquired goodwill and changes in the company's operating performance is significantly lower in the three years following the acquisition as compared to when the auditor does not issue an M&A KAM. To illustrate, we find that, at the mean amount of acquired goodwill in our sample, operating cash flows are 8.4 to 13.2 percent lower following the acquisition when the auditor issues an M&A KAM in the acquisition year compared to when the auditor does not.

These goodwill impairment and post-acquisition performance tests include control variables capturing firm and deal characteristics that proxy for materiality, information asymmetry, investors' market reaction to the initial M&A transaction announcement and other company and transaction characteristics that predict M&A failures (Lehn and Zhao 2006; Bens et al. 2012; Chen et al. 2018; Wangerin 2019; Ben-David et al. 2025). The results suggest that, when auditors disclose M&A KAMs, the expected benefits from managers' acquisition strategies are less likely to materialize and that goodwill reflects overpayment compared to when auditors do not disclose M&A KAMs. In addition, these results suggest that auditors' M&A KAM disclosures highlight risks about future post-acquisition performance that investors' initial expectations do not fully reflect.

In our third series of tests, we investigate whether auditors' M&A KAM disclosures contain information reflecting uncertainties about CEOs' acquisition strategies that ultimately lead to forced turnover. Our results show that the likelihood of forced CEO turnover is incrementally higher when there is poor post-acquisition performance *and* the auditor issues an M&A KAM in the acquisition year, compared to when the auditor does not. In contrast to the view that KAMs do not communicate useful information, the results suggest auditor M&A KAM disclosures reflect uncertainty around acquisition strategies that ultimately fail, which are incremental

to other indicators of failed acquisition strategies (Lehn and Zhao 2006; Bens et al. 2012). In terms of economic effects, we find that forced turnover for the CEO is 10 percent more likely when the company experiences poor post-acquisition performance and the auditor had issued an M&A KAM in the year of acquisition compared to when the auditor did not.

Considering these results collectively, we make several incremental contributions to the literature. First, we provide new insights on auditors' role in M&A transactions. Prior studies suggest that auditors influence M&A transaction outcomes by helping acquirers identify better acquisition targets (Louis 2005), reducing valuation uncertainty (De Franco et al. 2011; Allee and Wangerin 2018), performing internal control audits (Kravet et al. 2018), and acting as information intermediaries (Cai et al. 2016; Dhaliwal et al. 2016; Andreicovici et al. 2025). Our results add to this line of literature by revealing that M&A KAM disclosures communicate information about the uncertainty implicit in M&A transactions, complementing research showing that investors react negatively to *future* acquisition announcements after auditors issue M&A-related KAMs for a client's *previous* acquisitions (Abbott and Buslepp 2023).

Second, we contribute to research documenting the consequences of M&A transaction uncertainty and subsequent M&A performance (Erickson et al. 2012). We establish a link between uncertainty and the probability of M&A failure through our analysis of goodwill impairment and post-acquisition performance. Research by Ben-David et al. (2025) provides evidence that investors do not immediately distinguish successful from unsuccessful M&A transactions by showing that *ex ante* transaction announcement returns are poor predictors of future performance, including significant transaction failures that goodwill impairments reveal. By focusing on observable *ex post* outcomes instead of short-window market reactions that researchers measure before the acquisition is completed, our results show that KAM disclosures contain information about M&A transaction uncertainty that predicts subsequent M&A performance.

Third, we provide new evidence to the literature on CEO turnover following failed acquisitions. Lehn and Zhao (2006) show that CEO turnover is more likely when acquirers experience negative transaction announcement returns. Bens et al. (2012) find that CEOs are more likely to be replaced following failed acquisitions but can prolong their tenure in the short term by overstating post-acquisition earnings. We add to this literature by providing the first evidence linking uncertainty in M&A KAMs to *future* forced CEO turnover.

Our study also contributes to the emerging literature on expanded auditor reporting. Several studies find that KAM and CAM disclosures have no impact on audit quality, audit fees, or investor responses to the release of the financial statements (Bédard et al. 2019; Lennox et al. 2023).³ Other studies in this line of work make

³ The evidence in this line of research does not support the conclusion that KAMs and CAMs lack relevant information content. For example, the results of Lennox et al. (2023) suggest that investors are already aware of the financial reporting risks identified in auditor KAM disclosures and impound that information in equity prices. These findings suggest the KAMs provide relevant information by confirming investors' beliefs.

the implicit assumption that all KAMs/CAMs are equal by examining their determinants and consequences without differentiating between topical areas (Gutierrez et al. 2018). Reflecting on our study's incremental contributions, it is important to note our focus on the M&A setting and one specific type of KAM disclosure where uncertainty and accounting complexity are particularly high. Our results show that characteristics of the M&A transaction itself are among the most important predictors of an M&A KAM disclosure. Therefore, we include these characteristics in our KAM disclosure model, which distinguishes it from those in other studies that focus on characteristics of the client, individual auditor, or the audit firm as a whole (Ai 2023). By narrowing our focus to M&A KAMs, our research responds to the call in Minutti-Meza (2021) for research "determining whether common KAMs and CAMs topics have unique effects on complex aspects of financial reporting."

We proceed as follows. Section 2 reviews prior literature, along with articulating our conceptual model and hypotheses. Section 3 explains our methods, and Section 4 reports results. Section 5 concludes.

2 Literature review and hypotheses

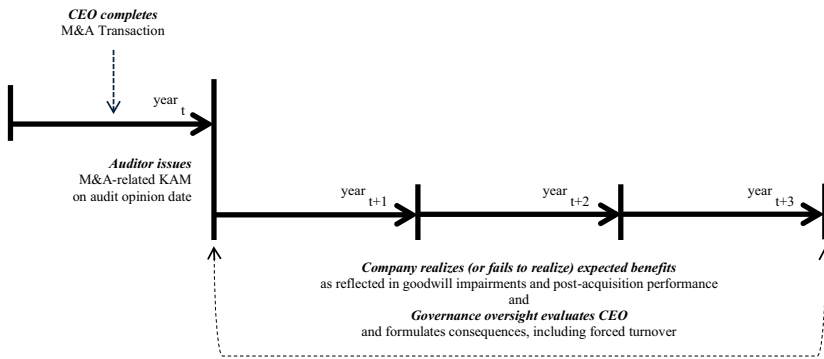
2.1 A conceptual model linking M&A and KAM reporting

Figure 1 Panel A outlines the timing of events linking M&A with KAM reporting, while Panel B lays out a conceptual model for our predictions. The timeline of events originates with the company's M&A transaction in year t . At the end of year t , the auditor decides whether to discuss the transaction and how to describe any relevant risks, uncertainties, and audit responses in an M&A KAM disclosure. Anticipated post-acquisition performance of the transaction depends on materiality, information asymmetry, and overpayment risk. In periods $t+1$, $t+2$, $t+3$, etc., managers will proceed with integration and operational activities, resulting in varying degrees of success or failure of their acquisition strategies. If the expected benefits of the acquisition fail to materialize, post-acquisition performance will suffer, and theory suggests that the board may consider incentives and disincentives for changes in top leadership, sometimes resulting in forced CEO turnover (Johnstone et al. 2011)

2.2 M&A transactions and uncertainty

The uncertainty arising from large, complex M&A transactions creates strong pressures on CEOs to achieve acquisition strategies that enhance shareholder value. Prior research suggests analysts and investors are unable to distinguish *ex ante* (i.e., at the time the transaction is announced) which M&A transactions will ultimately prove successful from those that will ultimately fail, and analysts have difficulty using financial statement information to assess post-acquisition performance (Erickson et al. 2012; Dickinson et al. 2016; Ben-David et al. 2025). Other research suggests CEOs gain private benefits through compensation following M&A (Jensen 1986; Grinstein and Hribar 2004; Harford et al. 2012), creating incentives to mask

Panel A: Timeline of events



Panel B: Conceptual model

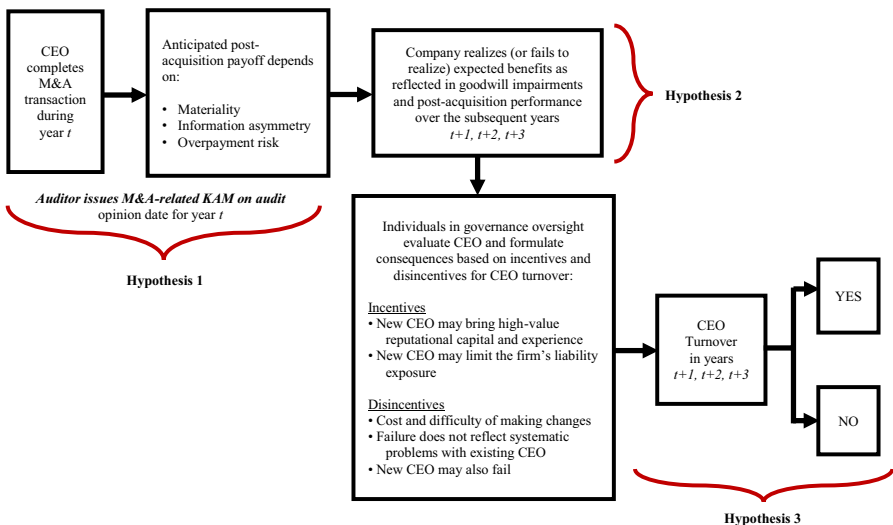


Fig. 1 Timeline and conceptual model

poor post-acquisition performance (Bens et al. 2012). This has motivated calls to improve disclosures that better enable investors to hold managers accountable for their acquisition decisions (CFA Institute 2020; IASB 2020, 2024).

Uncertainty about the future payoffs of M&A transactions arises from various sources. One source is the sheer size of the investment. The success of many acquisitions depends on achieving synergies through cost reductions and economies of scale. However, when acquiring a larger target company, it is more difficult and complex to successfully integrate that company's operations, employees, business processes, and internal control systems into the business than when acquiring a smaller target company. Consistent with this reasoning, research shows that larger

acquisitions are more prone to fail, as evidenced by negative abnormal stock returns in transactions involving larger targets (Moeller et al. 2004) and a higher probability of subsequent goodwill impairment in transactions with large allocations to goodwill (Hayn and Hughes 2006).

Another source of uncertainty arises from information asymmetries between the target and acquirer. Accurately assessing acquisition value depends on obtaining private information about the target's business, resources, obligations, and risks. Therefore, the acquirer and its CEO encounter adverse selection problems during the M&A transaction process, and in some cases, the extent of their information disadvantage can be severe (Datar et al. 2001). To alleviate this problem, the acquirer conducts due diligence, providing the opportunity to obtain and verify valuable private information directly from the target (Amel-Zadeh and Zhang 2015). Acquirers also experience lower post-acquisition profitability and recognize more goodwill impairments in transactions involving targets with higher information asymmetry as compared to those with lower information asymmetry (Chen et al. 2018; Wangerin 2019).

A third source of uncertainty derives from valuation errors and cognitive biases. The winner's curse hypothesis predicts that the winning bidder in an auction is more likely to overpay and that the amount of overpayment increases for assets with higher levels of valuation uncertainty (Thaler 1988). The winner's curse applies to M&A transactions and helps explain why CEOs overpay for acquisitions that ultimately fail (Ahern and Weston 2007; Brander and Egan 2017; De Bodt et al. 2018). Other explanations for M&A failure include the hubris hypothesis—the idea that managers may make valuation errors by over-estimating the value of the target and potential synergies (Roll 1986)—and the overconfidence hypothesis—the idea that managers have an inflated self-view of their abilities to achieve the objectives of M&A transactions (Malmendier and Tate 2008). Both of these explanations for M&A failure suggest that CEOs may overpay, especially in the presence of target company valuation uncertainty. Regardless of the reasons for overpayment, it results in recording greater amounts of acquired goodwill (Henning et al. 2000; Gu and Lev 2011; Li et al. 2011; Li and Sloan 2017).

The various sources of uncertainty surrounding M&A transactions necessitate that auditors identify associated risks of material misstatement and plan audit procedures to address those risks. Auditors are in a unique position to assess the sources of M&A-related uncertainty but have lacked a viable communication mechanism within the context of the traditional auditor reporting framework. KAM disclosures now provide such a mechanism.

2.3 M&A KAM disclosures

The UK Financial Reporting Council (FRC) initiated expanded auditor reporting with risk of material misstatement (RMM) disclosures in the audit reports of UK companies with a premium listing on the London Stock Exchange (LSE) Main Market with fiscal year-ends on or after September 2013 (FRC 2013). In June 2016, the FRC revised its standards to conform with IAASB KAM reporting requirements. The revised FRC rules became effective for all companies listed on the LSE, including the Main Market and Alternative Investment Market, with fiscal year-ends on or after June 2017, with

early adoption permitted (FRC 2016a, 2016b).⁴ KAMs reveal the most significant matters in the audit—they often relate to areas involving subjective judgments and high estimation uncertainty that may require the use of specialists—and include descriptions of procedures that auditors employ to respond to these matters (FRC 2016b). Regulators intended that auditors use KAMs to describe the most significant risks, transactions, events, and/or estimates they encountered during each engagement (FRC 2016b).

There are several similarities and differences that exist in UK KAM versus U.S. CAM reporting requirements. Both the FRC and PCAOB shared the objective that KAMs and CAMs would benefit users via enhanced transparency about the assurance process. However, KAMs are defined as matters that, in the auditor's professional judgment, were of most significance in the audit, while CAMs must meet two specific criteria: (1) they must relate to accounts or disclosures that are material to the financial statements and (2) involve especially challenging, subjective, or complex auditor judgment. KAM reporting applies to both publicly traded and privately held companies following International Standards on Auditing, whereas CAM reporting applies only to audits of U.S. public companies. However, auditors conducting U.S. public company engagements face heightened legal scrutiny compared to auditors elsewhere. For example, the U.S. audit litigation environment considers a broad exchange of information via depositions and extensive documentation requests, whereas the United Kingdom features a narrower discovery process.

Under IFRS 3, *Business Combinations*, the acquirer must recognize all identifiable assets that it purchases and all liabilities that it assumes as of the acquisition date. Furthermore, the acquirer must measure each identifiable asset and liability at fair value and allocate the remaining excess purchase price to goodwill. In many acquisitions, the most significant acquired assets are identifiable intangibles for which there is no active market. Therefore, fair value measurement of acquired intangibles is generally based on valuation models that require unverifiable estimates that depend on highly subjective assumptions (e.g., future growth projections in sales and margins, expected customer attrition, or discount rates). Due to these uncertainties, estimates, and assumptions, M&A transactions heighten audit risk.

Auditors therefore evaluate management's identification of the acquired identifiable assets and liabilities, fair value estimates, and the residual allocation to goodwill with a particular focus on underlying valuation assumptions that are key inputs to fair value estimates of acquired intangibles. Audit guidance from the IAASB for fair value estimates requires that auditors gain an understanding of management's valuation approach and evaluate the reasonableness of significant assumptions (IAASB 2018). In following these standards, auditors compare assumptions against external benchmarks, historical information, and projections in other internal reports prepared by management (e.g., due diligence reports, internal forecasts, or reports to

⁴ RMMs from the 2013 standards and KAMs from the 2016 standards are largely similar, with the more recent rules providing an enhanced framework to guide the auditor in discussing KAMs (Minutti-Meza 2021). For simplicity, we use the term "KAMs" throughout the paper, but in doing so we are referring to both RMMs and KAMs.

the board of directors) and utilize valuation specialists.⁵ Consistent with the requirements of the relevant audit guidance, auditor M&A KAM disclosures typically discuss management's identification of the acquired net assets, fair value estimates for acquired intangible assets, the significant assumptions used by management to support their fair value estimates, and the residual allocation of the purchase consideration to goodwill. Appendix 1 includes excerpts from two M&A KAMs that serve as representative examples. As one example, we include below relevant excerpts from The Daily Mail FYE 2015 M&A KAM:

The Group continues to make significant investment in business acquisitions. During the year the Group made a number of acquisitions ... IFRS 3 'Business combinations' requires the fair value of consideration paid to be allocated firstly to the fair value of net assets acquired, then to the fair value of separately identifiable intangible assets and finally to goodwill, which involves a number of estimates and judgements.

In explaining its responses later in the KAM, the auditor notes:

We focused our work on the acquisitions and disposals during the year with the largest value of purchase consideration and those where the terms were not straightforward... assessed the appropriateness of the directors' identification of intangible assets acquired ... obtained the directors' calculation of the fair value of intangible assets acquired, and corroborated the appropriateness of the assumptions over royalty rates, levels of customer attrition, useful economic lives and forecast cash flows to external benchmarks, historical performance and the acquisition plans.

As another example, RWS Holding's FYE 2017 M&A KAM describes a material acquisition of LUZ, Inc. with valuation uncertainty related to goodwill and intangibles:

The Group acquired LUZ Inc. in February 2017 for a cash consideration of US\$82.5 million. Accounting for the acquisition required a fair value exercise to assess the assets and liabilities acquired, including valuing any separately identifiable intangible assets with the residual balance recognized as goodwill. The valuation of identified intangibles can be a subjective process and as such was an area of focus for us.

⁵ In addition to ISA 540, several additional references are applicable. The IAASB standards discuss risk assessment procedures in general (IAASB 2024, International Standard on Auditing 315, see especially paragraphs 28–37). ISA 315 requires that auditors identify risks of material misstatement and evaluate how these risks may pervasively affect the financial statements, focusing particularly on inherent and control risk; the auditor will then gather and evaluate audit evidence and assemble relevant documentation. As it relates to acquisitions and consolidations, ISA 600 provides details on a host of issues, including understanding the organizational structure of the combined entity (Para. A93), evaluating the system of internal control (Appendix 2, Para. 6), and evaluating the consolidation process, particularly with respect to considering fair value assessments and goodwill and changes therein (Appendix 2, Para. 6) (IAASB 2022). IOSCO (2023) makes specific recommendations for the audit of goodwill (e.g., challenging management's assumptions and disclosures, Section 5.3).

In responding to the valuation uncertainty about goodwill, the auditor states that it:

Assessed the completeness and quantum of intangible assets identified by management against our own expectations, formed from review of the due diligence reports prepared by management's professional advisors... Assessed the work performed on the purchase price allocation by utilizing our in-house specialists to evaluate management's valuation of the identified assets. Specifically, we reviewed the methodology adopted, compared the assumptions made on attrition and recoverability with historical patterns in the business to verify that assumptions were reasonable... Corroborated the value of intangibles by performing an overall sense-check of the level of residual goodwill arising on the transaction.

2.4 Hypotheses

KAMs focus users' attention on auditors' concerns about whether managers adequately identified the assets the company acquired and the liabilities the company assumed, incorporated the relative complexity of fair value estimates in the transaction price, and made accurate projections about uncertain future events. As the previous examples highlight, auditors respond to this uncertainty by increasing audit effort, for example, assessing the identification of the acquired net assets, management's fair value estimates, and the reasonableness of key valuation assumptions for the acquisition that are relevant predictors of future performance. When M&A transactions are subject to more economic uncertainty, we expect there will be accompanying accounting uncertainty, which auditors will describe via M&A KAM disclosures that identify these risks and explain procedures the auditor performed to address them.

Under the expanded auditor reporting model, KAMs provide auditors the opportunity to communicate information about their judgments, including the level of uncertainty underlying M&A transactions. Taken together, this discussion suggests the following hypotheses:

H1a: *There is a positive association between M&A KAM disclosures in the year of acquisition and the materiality of underlying M&A transactions.*

H1b: *There is a positive association between M&A KAM disclosures in the year of acquisition and the information asymmetry underlying M&A transactions.*

H1c: *There is a positive association between M&A KAM disclosures in the year of acquisition and the risk of overpayment in M&A transactions.*

Our conceptual model predicts that M&A transactions with more uncertainty will be associated with weaker post-acquisition performance, for example, goodwill impairments and lower operating cash flows, and a higher likelihood of forced CEO turnover, compared to M&A transactions with less uncertainty. The underlying intuition for these predictions is that, when more uncertainty exists, the future payoff to a CEO's M&A decision has a wider range of future outcomes, including transaction

failures. Consistent with the winner's curse and the hubris and the overconfidence hypotheses, prior literature shows that CEOs often overestimate their ability to successfully execute their M&A investment strategies, leading to overpayment, poor post-acquisition performance, and goodwill impairment (Roll 1986; Malmendier and Tate 2008; De Bodt et al. 2018; Chung and Hribar 2021).

The amount of goodwill recognized in the transaction has particularly important implications for predicting future improvements or declines in post-acquisition performance. On the one hand, goodwill captures several sources of future benefits arising from the acquisition. Measured as the remaining amount of the purchase price after deducting the fair value of the target's net identifiable assets under U.S. GAAP and IFRS, the residual amount recognized in goodwill reflects unidentifiable assets (i.e., assets that cannot be recognized separately), such as the target firm's human capital, its competitive advantages in the market, and expected synergies from the acquisition (Johnson and Petrone 1998; Henning et al. 2000; Linsmeier et al. 2024). Therefore, acquisitions with larger amounts of goodwill may predict improvements in future performance if the company realizes the expected benefits from the acquisition.

On the other hand, goodwill can also reflect overpayment if management overestimates the value of the target firm and its ability to successfully achieve its acquisition strategies (Johnson and Petrone 1998; Hayn and Hughes 2006; Li et al. 2011; Li and Sloan 2017; Chen et al. 2018; Wangerin 2019). To the extent that managers' expected future benefits of the acquisition are more optimistic than the underlying assumptions used in their fair value estimates for identifiable intangibles (e.g., future sales growth rates or customer attrition), we expect lower allocations to intangibles and higher allocations to goodwill. Moreover, greater uncertainty about the future benefits of the acquisition should produce heightened auditor scrutiny of management's fair value assumptions that shift allocations from acquired intangibles to goodwill when the risk of overpayment is higher. In this case, acquisitions with larger amounts of goodwill may predict declines in future performance and pose a greater risk of future impairment.

Our second set of hypotheses examines how goodwill predicts future post-acquisition performance realizations, conditional on whether the auditor issues an M&A KAM in the acquisition year. If auditors communicate the level of uncertainty surrounding M&A transactions in KAM disclosures, we expect M&A KAMs will reflect uncertainty about whether expected synergies and other sources of benefits in acquired goodwill will be realized. Therefore, we expect to observe an increase in future goodwill impairments and that the goodwill recognized in the transaction will exhibit a weaker association with future performance when the auditor issues an M&A KAM. This discussion leads to the following hypotheses:

H2a: *There is a positive association between acquisition-year M&A KAM disclosures and future goodwill impairments.*

H2b: *The association between goodwill and future post-acquisition performance will be less positive when the auditor issues an M&A KAM in the year of acquisition.*

To the extent that managers fail to successfully execute the strategies motivating M&A transactions, declining post-acquisition profitability will eventually be revealed in acquirers' post-acquisition financial statements. Our conceptual model predicts that the board of directors will react to poor post-acquisition performance by implementing various changes in an attempt to restore equilibrium (Agrawal et al. 1999), including those involving leadership (Larcker et al. 2007; Johnstone et al. 2011; Hoitash et al. 2012). Incentives motivating leadership changes include hiring a new CEO to bring high-value reputational capital and experience in remedying underlying causes of the M&A failure. A new CEO may also limit the company's liability because that person's presence indicates a willingness to work toward remedying underlying problems (Agrawal and Cooper 2017). Disincentives include separation costs and difficulties hiring a replacement CEO, or a belief that the M&A failure is not the fault of the existing CEO (Agrawal et al. 1999). Depending on the balance of these incentives and disincentives, the board will decide on a recovery plan, possibly including firing the CEO. If auditors communicate the level of uncertainty surrounding M&A transactions in KAM disclosures, we expect M&A KAMs to predict a higher probability of forced CEO turnover following poor post-acquisition performance, leading to the following hypothesis:

H3: *The association between poor post-acquisition performance and CEO turnover will be more positive when the auditor issues an M&A KAM in the year of acquisition.*

Our hypotheses are not without tension. Some research reveals that KAMs are positively associated with audit fees and quality (Li et al. 2019; Reid et al. 2019), reflect companies' levels of financial distress (Camacho-Miñano et al. 2024), influence debt contracting (Porumb et al. 2021), and yield investor reactions around annual report filings (Goh et al. 2024). Additional evidence suggests that the communicative value of audit reports has increased after KAM reporting (Seebeck and Kaya 2023) and that KAMs affect managers' decision-making, financial reporting, and disclosure choices (Lynch et al. 2023; Andreicovici et al. 2025; Drake et al. 2024).

Other evidence in prior research concerning KAM and CAM reporting raises questions about whether auditors will make M&A KAM disclosures to identify and communicate risks to investors and, if so, whether those disclosures are useful for predicting the future success or failure of CEOs' acquisition strategies. For example, Gutierrez et al. (2018), Bédard et al. (2019), Burke et al. (2023), and Liao et al. (2024) report that expanded reporting regimes in China, France, the United Kingdom, and the United States are not associated with investor reactions, audit fees, audit report delay, audit quality, or a combination of these. This research questions the relevance of KAM/CAM reporting and suggests that auditors may be using boilerplate language in response to litigation risk rather than as a communication channel to inform investors.

3 Methods

3.1 Sample description

Our sample consists of all companies listed on the LSE, including the Main Market and Alternative Investment Market, which completed 2,460 M&A transactions with fiscal years ending from September 2013 through December 2020 (Table 1 Panel A). Expanded auditor reporting requirements became effective in the United Kingdom for companies with a premium listing on the Main Market with fiscal year-ends on or after September 2013, which is when we begin our sample. We conclude the sample in December 2020 so that we can populate three-year forward-looking variables with the most recent financial data available.⁶ We obtain M&A transaction data from the SDC Global Mergers and Acquisitions database, KAM and auditor-related data from Audit Analytics Europe, CEO data from BoardEx UK, financial data from FactSet International, and stock return data from Refinitiv Datastream.⁷

Because some acquirers complete multiple M&A transactions in a fiscal year, our sample size for the KAM disclosure model (H1) begins with 1,368 company-year observations.⁸ Following prior research, we eliminate 77 observations with transaction values less than 1% of the acquirer's total assets (Cai et al. 2016; Chen et al. 2018; Kravet et al. 2018). We lose another 266 observations because of missing financial or auditor-related data needed to create our variables, which yields a sample of 1,025 company-year observations for testing H1a, H1b, and H1c. Sample sizes for our goodwill impairment, post-acquisition performance, and CEO turnover models are further reduced by additional financial or auditor-related missing data (yielding N = 580 to 953). See Table 1 Panel B for a breakout by industry, and see Table 1, Panel C for a breakout by year and market. Our sample covers a wide array of industries and is relatively evenly split between Main Market and Alternative Investment Market and other LSE companies.

⁶ We include all companies listed on the LSE for the full sample period of September 2013 through December 2020, as early adoption of expanded auditor reporting requirements by non-main market companies was permitted. Our sample includes 34 observations of non-main market companies that had M&A KAMs included in their audit reports prior to mandatory adoption. In robustness tests, we reperform our analyses to rule out the potential alternative explanation that our findings are driven by the effects of voluntary adoption. We discuss these results in Sect. 4.6.

⁷ We use Refinitiv Datastream to obtain stock return data, which results in the loss of approximately 20–45 observations across our post-acquisition performance and CEO turnover models due to missing data. Our results are robust when excluding controls for stock returns.

⁸ Frequency counts (percentages) for the number of transactions by company-year are as follows: 852 (62%) = one transaction, 291 (21%) = two transactions, 108 (8%) = three transactions, 117 (9%) = four or more transactions.

Table 1 Sample description

Panel A: Summary of sample construction					
	M&A Transactions	[H1a,b,c] KAM Disclosure Model	[H2a] Goodwill Impairment Model	[H2b] Operating Cash Flow Models Year t + 1 Year t + 2	[H3] CEO Turnover Model Year t + 3
London Stock Exchange companies with completed M&A transactions, September 2013–December 2020	2,460 [completed transactions]	1,368 [company-year observations]			
Less: Observations not meeting sample criteria (i.e., M&A transactions ≤ 1% of total assets)	–107	–77			
Less: Observations with missing data for variables included in the KAM disclosure model	–598	–266			
Number of company-year observations for the KAM disclosure model	1,755	1,025	1,025	1,025	1,025
Less: Observations with missing data for variables included in the consequence models			–129	–72	–445
Number of company-year observations for the consequence models			896	953	580
Panel B: Sample composition by industry					
Industry (one-digit SIC code)	# of M&A Transactions for Acquirers	# of M&A Transactions for Targets			
0—Agriculture, Forestry, and Fishing	5	17			
1—Mining and Construction	75	89			
2—Consumer Nondurables Manufacturing	179	145			
3—Consumer Durables Manufacturing	250	232			

Table 1 (continued)

4—Transportation and Public Utilities	106	159	
5—Wholesale and Retail Trade	87	120	
6—Finance, Insurance, and Real Estate	509	392	
7—Business Equipment Services	325	385	
8—Other Services	217	212	
9—Public Administration	2	4	
Total	1,755	1,755	
Panel C: Sample composition by year and market			
Year	# of M&A Transactions for Main Market (MM) Companies	# of M&A Transactions for Alternative Investment Market (AIM) & Other LSE Companies	Total # of M&A Transactions
2013	87	50	137
2014	135	122	257
2015	159	120	279
2016	104	126	230
2017	119	110	229
2018	119	104	223
2019	102	111	213
2020	81	106	187
Total	906	849	1,755

This table explains sample construction and composition. The sample includes London Stock Exchange companies with completed M&A transactions from September 2013 – December 2020. We summarize M&A transactions at the company-year level for purposes of analysis. We obtain M&A transaction data from the SDC Global Mergers and Acquisitions database, KAM and auditor-related data from Audit Analytics Europe, CEO data from BoardEx UK, financial data from FactSet International, and stock return data from Refinitiv Datastream.

3.2 Analysis of M&A KAM disclosures

We test H1a, H1b, and H1c by estimating the following M&A KAM disclosure model:

$$M\&A\ KAM = \beta_0 + \beta_1 Materiality + \beta_2 Information\ Asymmetry + \beta_3 Overpayment\ Risk + Controls + \gamma + \varepsilon \quad (1)$$

We estimate a linear probability model for the dependent variable *M&A_KAM*, which equals one if the audit report includes an auditor KAM disclosure related to accounting for business combinations in the acquisition completion year and zero otherwise.⁹

We estimate the likelihood that auditors will make M&A KAM disclosures based upon materiality, information asymmetry, and overpayment risk. To test H1a, we proxy for materiality using the variable *RELATIVE_SIZE*, which equals the purchase price scaled by the acquirers' market value of equity at the end of year $t - 1$, aggregated by company-year. To test H1b, we create two variables to capture information asymmetry. First, *CROSS_INDUSTRY* is equal to the number of cross-industry deals (i.e., when the two-digit SIC codes of the target and acquirer differ) completed for each company-year observation (Datar et al. 2001). Second, *FOREIGN_TARGET* is equal to the number of foreign target deals (i.e., when the target company is headquartered outside the United Kingdom) completed for each company-year observation (Kohers and Ang 2000). To test H1c, we proxy for overpayment risk with the variable *GOODWILL*, which equals the change in goodwill from year $t - 1$ to year t (plus impairment in year t) scaled by the acquiring company's total assets at the end of year $t - 1$.

We draw from models in prior literature on KAMs/CAMs (Sierra-García et al. 2019; Burke et al. 2023) and include control variables for the following auditor-specific characteristics: *BIG4*, *ΔAUDITOR*, and *RESTATE*. *BIG4* is an indicator variable equal to one if PwC, EY, Deloitte, or KPMG performed the audit and zero otherwise. *ΔAUDITOR* is an indicator variable equal to one if the auditor changed in the current year and zero otherwise. *RESTATE* is an indicator variable equal to one if the issuer disclosed a restatement in the current or prior two fiscal years and zero otherwise. We also control for company-specific characteristics related to complexity and performance, including *ASSETS*, *BTM*, *SALES_GROWTH*, *FOREIGN_SALES*, *LEVERAGE*, *ROA*, and *RESTRUCTURING* of the acquirer. For brevity, we describe definitions for these control variables in Appendix 2. In this and all other analyses, we winsorize all continuous variables at the 1st and 99th percentiles, include year and industry fixed effects (denoted γ) based on one-digit SIC codes, and estimate clustered standard errors by company.¹⁰

⁹ We identify KAMs related to business combinations using the subject area mapping in Audit Analytics (identified as topic key 4). We also manually reviewed KAMs mapped to significant non-routine transactions (topic key 54) and consolidations (topic key 62), identifying 12 KAMs that discuss aspects of business combinations (typically along with additional matters such as disposals). Our results are robust to coding these other KAM topics as M&A KAMs.

¹⁰ If we employ two-digit SIC codes, the sample size declines due to perfect prediction of the dependent variable; therefore, we employ one-digit SIC codes to minimize loss of data.

3.3 M&A KAM disclosures and post-acquisition performance

We test H2a by estimating the following goodwill impairment model:

$$\text{Goodwill Impairment Variable} = \beta_0 + \beta_1 \text{M\&A KAM} + \text{Controls} + \gamma + \varepsilon \quad (2a)$$

We use three different dependent variables to test the prediction that acquisition-year M&A KAMs predict future goodwill impairments. First, we estimate a linear probability model where the dependent variable is *GW_IMPAIR*, which is an indicator variable equal to one when the company recognizes a goodwill impairment in year $t+1$, $t+2$, or $t+3$ following the acquisition and zero otherwise. Second, we estimate an OLS regression model where the dependent variable is *GW_IMPAIR_AMT*, which is the cumulative amount of goodwill impairment losses (scaled by sales) recognized over the three years after the acquisition. Third, we estimate a linear probability model where the dependent variable is *GW_KAM_3YR*, an indicator variable equal to one if the audit report includes an auditor KAM disclosure relating to goodwill within the three years following the acquisition and zero otherwise.

If M&A KAMs reflect greater uncertainty about the ultimate success or failure of management's acquisition strategies and a higher likelihood of overpayment, we expect that the β_1 coefficient will be positive and significant. We use the alternative dependent variables to make inferences about whether M&A KAMs predict the likelihood, amount, and risk of future goodwill impairment. Although testing the association between M&A KAMs and future goodwill impairment is a direct test of whether these auditor disclosures predict poor post-acquisition performance, the dependent variables are noisy proxies for failed acquisition strategies. The literature shows that managers exploit subjectivity in accounting guidance for goodwill impairment to manipulate the amount and timing of loss recognition (Ramanna and Watts 2012; Li and Sloan 2017; Ayres et al. 2019). To the extent that managers minimize, delay, or avoid recognizing goodwill impairment losses, this biases against our ability to detect the predicted effects. These effects may be mitigated to some extent in our analysis where we use future goodwill KAMs as the dependent variable. In circumstances where managers exploit the inherent subjectivity in the goodwill impairment accounting standards (without violating IFRS), we expect auditors' future goodwill KAM disclosures to identify areas requiring higher levels of complex judgment and a heightened risk of future impairment.

We test H2b by estimating the following post-acquisition performance model:

$$\text{Post-Acquisition Performance} = \beta_0 + \beta_1 \text{M\&A KAM} + \beta_2 \text{Goodwill} + \beta_3 \text{M\&A KAM} * \text{Goodwill} + \text{Controls} + \gamma + \varepsilon \quad (2b)$$

We estimate equation [2b] using OLS regression models in which the dependent variable captures changes in post-acquisition profitability in each of the first three years after the acquisition: ΔOCF_{t+1} , ΔOCF_{t+2} , and ΔOCF_{t+3} . We calculate ΔOCF as the change in operating performance from year $t-1$ to year $t+n$, where operating performance equals operating cash flows divided by sales. We utilize operating cash flows to measure post-acquisition performance as opposed to income, because

operating cash flows are less impacted by accounting effects arising from the acquisition method of accounting, such as the effects of depreciation and amortization (Powell and Stark 2005; Dickinson et al. 2016).¹¹ In addition, this analysis complements our tests of goodwill impairment, as it captures variation in both positive and negative post-acquisition performance and does not rely on whether management records a future impairment.

The intuition for this test is that the goodwill balance reflects the expected future benefits of the acquisition implied by the purchase price and the fair value assigned to the acquired net identifiable assets. To the extent that goodwill predicts realizations of expected benefits from the acquisition, this will be reflected in higher post-acquisition operating performance. Therefore, we expect that the coefficient on *GOODWILL* (β_2) will be positive. If M&A KAM disclosures reflect greater uncertainty about the expected future benefits of the acquisition and a higher likelihood of overpayment, we expect that goodwill will predict lower post-acquisition operating performance. In accordance with H2b, we predict that the coefficient for the *M&A_KAM*GOODWILL* interaction (β_3) will be negative.¹²

Following prior literature (e.g., Chen et al. 2018; Wangerin 2019), we control for company characteristics in the models related to goodwill impairment and post-acquisition performance, including *ASSETS*, *BTM*, and *LEVERAGE*. We also control for investors' market reaction to the M&A transaction announcement (*AB_ANN_RET*) (Lehn and Zhao 2006). In addition, we include controls for the form of payment (*ALL_STOCK* and *ALL_CASH*) and the total number of M&A transactions completed by the acquirer in year t (*#ACQUISITIONS*). We further control for whether the acquirer's auditor issued a goodwill KAM prior to year t (*GW_KAM*) and the total number of KAMs covering topics other than business combinations or goodwill included in the audit report (*#OTHER_KAMS*). Last, we include proxies to control for materiality and information asymmetry using the variables from the M&A KAM disclosure model (*RELATIVE_SIZE*, *CROSS_INDUSTRY*, and *FOREIGN_TARGET*). We include these controls to rule out alternative explanations that goodwill impairment and post-acquisition performance are explained by underlying M&A transaction characteristics that are correlated with M&A KAM disclosures.

3.4 M&A KAM disclosures and forced CEO turnover

We test H3 with a forced CEO turnover model using a linear probability model:

$$CEO_TURN = \beta_0 + \beta_1 M\&A_KAM + \beta_2 BAD_PERFORM + \beta_3 M\&A_KAM * BAD_PERFORM + Controls + \gamma + \varepsilon \quad (3)$$

¹¹ Our results are robust to measuring the change in operating performance using total assets as an alternative scalar and measuring the change in operating performance using as net income scaled by total assets.

¹² Refer to Appendix 3 for a numerical illustration supporting the predicted negative coefficient for β_3 .

We follow prior literature and classify CEO turnover as forced using an algorithm based on a combination of press releases and CEO age (Parrino 1997; Peters and Wagner 2014).¹³ We classify CEO turnover as forced when the press release states that the CEO is fired, forced out, or retires or resigns due to policy differences. If the CEO is less than 60 years of age, we classify CEO turnover as forced if the press release does not report the reason for turnover to be death, poor health, or acceptance of another position, or the article reports that the CEO is retiring but the company does not announce the retirement date at least six months before departure. In addition, we classify any instances of CEO turnover in which the CEO either remains on the board or within management as unforced turnover. Based on this classification, we set *CEO_TURN* equal to one if the CEO was forced out within the three years following the M&A transaction and zero otherwise.

If uncertainty about future performance reflected in M&A KAM disclosures includes a wider range of possible outcomes, including significant deal failures, we expect a positive coefficient on the *M&A_KAM* variable. The variable denoted *BAD_PERFORM* represents poor post-acquisition performance, which we capture with two alternative independent variables. First, we use *ΔOCF_NEGATIVE*, an indicator variable equal to one if the average change in operating performance over the three years following the acquisition is negative and zero otherwise. Second, we use the variable *GW_IMPAIR_AMT* as previously defined in our goodwill impairment analysis.¹⁴ Although both *ΔOCF_NEGATIVE* and *GW_IMPAIR_AMT* should reliably capture poor performance, it is unclear whether poor performance in any given company is attributable to the CEO's acquisition decisions or other unrelated factors. Therefore, we interact *M&A_KAM* with both variables to make inferences about decisions to terminate the CEO when the board determines poor performance related to failed acquisition strategies. H3 predicts a positive coefficient for β_3 , consistent with poor post-acquisition performance predicting a higher probability of future CEO turnover when the auditor issues an M&A KAM in the acquisition year as compared to when the auditor does not.

We control for materiality, information asymmetry, and overpayment risk consistent with the variables we include in our M&A KAM disclosure model: *RELATIVE_SIZE*, *CROSS_INDUSTRY*, *FOREIGN_TARGET*, and *GOODWILL*. We include these controls to rule out alternative explanations that forced CEO turnover is explained by underlying M&A transaction characteristics that are correlated with

¹³ Our sample of company-years with M&A transactions includes 29 instances of forced CEO turnover in year $t+1$ (35 percent), 26 instances in year $t+2$ (32 percent), and 27 instances in year $t+3$ (33 percent); these forced turnover rates are somewhat higher than those that Parrino (1997) and Peters and Wagner (2014) report, which are 13 percent and 24 percent, respectively. Further, these turnover rates are similar to Lehn and Zhao (2006), who find that 27 percent of CEOs are replaced within five years of an acquisition through internal governance mechanisms.

¹⁴ We elect to use *GW_IMPAIR_AMT* in our analysis of forced CEO turnover because this variable captures both the incidence and magnitude of goodwill impairment loss recognition. This allows us to conduct more powerful tests of H3 than using the alternative proxies from our goodwill impairment analysis *GW_IMPAIR* and *GW_KAM_3YR*.

M&A KAM disclosures. We also control for investors' market reaction to the M&A transaction announcement (*AB_ANN_RET*), as previously defined (Lehn and Zhao 2006). Consistent with our post-acquisition performance models and prior literature examining CEO turnover (Johnstone et al. 2011; Bens et al. 2012), we control for company-specific characteristics relating to performance and complexity, including acquirer *ROA*, ΔROA_{t-1} , $\Delta SALES_{t-1}$, ΔRET_{t-1} , *ASSETS*, *BTM*, *LEVERAGE*, *LOSS*, and *RESTATE*. We measure these control variables related to performance in the years leading up to the acquisition. We further control for the total number of completed M&A transactions by the acquirer in the current year (*#ACQUISITIONS*), whether the acquirer's auditor issued a goodwill KAM prior to year *t* (*GW_KAM*), and the total number of KAMs covering topics other than business combinations or goodwill included in the audit report (*#OTHER_KAMS*). In addition, we control for CEO characteristics, including the number of years the CEO has held the position (*CEO_TENURE*) as well as whether the CEO is also the chair of the board of directors (*CEO_CHAIR*).

4 Results

4.1 Descriptive statistics

Table 2 Panel A includes descriptive statistics for the variables used in our analysis of M&A KAM disclosures (Model [1]). We find that 35 percent of company-year observations with M&A transactions have KAM disclosures related to accounting for business combinations.¹⁵ The mean *RELATIVE_SIZE* of each M&A transaction is 40 percent of the acquiring company's market value of equity, and the mean number of *CROSS_INDUSTRY* deals is just short of one (0.93 deals per company year); neither of these characteristics differs significantly for companies with versus without an M&A KAM. The mean *FOREIGN_TARGET* is 0.61 deals, and this amount is higher for companies with (0.85) versus those without an M&A KAM (0.48) (p-value < 0.01; two-tailed). The mean amount of acquired *GOODWILL* equals 11 percent of acquiring companies' total assets. The mean amount of *GOODWILL* for companies with an M&A KAM is not significantly different from those without an M&A KAM.

Table 2 Panel B includes descriptive statistics for variables used in our analyses of future goodwill impairment, post-acquisition performance, and CEO turnover (Models [2a], [2b], and [3], respectively). Across all firm-year observations in the sample, 26 percent recognize a goodwill impairment within three years of the acquisition (*GW_IMPAIR*), and the mean impairment is 3 percent of sales. The mean *GW_IMPAIR* (*GW_IMPAIR_AMT*) is 0.29 (0.04) for observations where the auditor reported an M&A KAM in the acquisition year, which exceeds the mean *GW_IMPAIR* (*GW_IMPAIR_AMT*) of 0.24 (0.03) for observations without M&A KAMs.

¹⁵ Our sample includes 135 RMM disclosures (for audit reports issued for fiscal years ending before June 17, 2017) and 227 KAM disclosures (for audit reports issued for fiscal years ending on or after June 17, 2017), which we refer to collectively as KAM disclosures.

Table 2 (continued)

Panel B: Goodwill impairment, operating cash flows, and CEO turnover model variables

Variable	All Company-Years				Company-Years with M&A KAMs				Company-Years without M&A KAMs				t/chi-square statistic	p-value	
	N	Mean	SD	Median	N	Mean	SD	Median	N	Mean	SD	Median			
Dependent Variables:															
<i>GW_IMPAIR</i>	896	0.26	0.44	0.00	314	0.29	0.45	0.00	582	0.24	0.43	0.00	1.51	0.07	
<i>GW_IMPAIR_AMT</i>	896	0.03	0.11	0.00	314	0.04	0.12	0.00	582	0.03	0.10	0.00	1.61	0.05	
<i>GW_KAM_3YR</i>	896	0.50	0.50	1.00	314	0.66	0.47	1.00	582	0.42	0.49	0.00	6.98	< 0.01	
<i>ΔOCFt+1</i>	953	0.12	0.87	0.01	339	0.06	0.70	0.00	614	0.15	0.94	0.01	-1.54	0.06	
<i>ΔOCFt+2</i>	919	0.12	0.97	0.01	326	0.05	0.72	0.00	593	0.16	1.08	0.01	-1.66	0.05	
<i>ΔOCFt+3</i>	815	0.09	0.63	0.01	278	0.03	0.42	0.00	537	0.11	0.71	0.01	-1.82	0.03	
<i>CEO_TURN</i>	580	0.14	0.35	0.00	209	0.16	0.37	0.00	371	0.13	0.34	0.00	0.73	0.39	
Control Variables:															
<i>AB_ANV_RET</i>	953	0.03	0.08	0.02	339	0.03	0.08	0.02	614	0.03	0.07	0.02	1.08	0.28	
<i>ASSETS</i>	953	4.97	2.02	4.74	339	5.64	1.90	5.49	614	4.61	1.99	4.41	7.78	< 0.01	
<i>BTM</i>	953	0.59	0.46	0.44	339	0.50	0.42	0.36	614	0.64	0.47	0.51	-4.38	< 0.01	
<i>LEVERAGE</i>	953	0.17	0.16	0.14	339	0.18	0.16	0.16	614	0.17	0.16	0.14	1.38	0.17	
<i>ALL_STOCK</i>	953	0.03	0.17	0.00	339	0.03	0.16	0.00	614	0.04	0.18	0.00	-0.76	0.45	
<i>ALL_CASH</i>	953	0.23	0.41	0.00	339	0.21	0.39	0.00	614	0.24	0.42	0.00	-1.07	0.28	
<i>#ACQUISITIONS</i>	953	1.72	1.38	1.00	339	1.86	1.64	1.00	614	1.64	1.20	1.00	2.38	0.02	
<i>#OTHER_KAMS</i>	953	1.32	1.75	0.00	339	1.78	1.85	2.00	614	1.06	1.64	0.00	6.18	< 0.01	
<i>GW_KAM</i>	953	0.23	0.42	0.00	339	0.38	0.49	0.00	614	0.14	0.35	0.00	9.02	< 0.01	
<i>ROA</i>	580	0.04	0.13	0.05	209	0.05	0.10	0.06	371	0.03	0.14	0.05	2.16	0.03	
<i>ΔROAt-1</i>	580	0.01	0.09	0.00	209	0.01	0.06	0.00	371	0.01	0.11	0.00	-0.99	0.71	

Table 2 (continued)

<i>ΔSALES_{t-1}</i>	580	-0.02	0.15	0.00	209	-0.01	0.13	0.00	371	-0.03	0.17	0.00	0.92	0.36
<i>ΔRET_{t-1}</i>	580	0.43	1.05	0.20	209	0.38	0.80	0.20	371	0.46	1.17	0.20	-0.91	0.36
<i>LOSS</i>	580	0.16	0.36	0.00	209	0.11	0.31	0.00	371	0.18	0.39	0.00	5.42	0.02
<i>RESTATE</i>	580	0.04	0.21	0.00	209	0.06	0.23	0.00	371	0.04	0.19	0.00	1.21	0.27
<i>CEO_TENURE</i>	580	5.13	4.95	3.52	209	4.98	4.81	3.24	371	5.20	5.04	3.77	-0.51	0.61
<i>CEO_CHAIR</i>	580	0.02	0.15	0.00	209	0.01	0.12	0.00	371	0.03	0.16	0.00	0.97	0.33

This table describes data for the variables included in our hypothesis tests. Descriptive statistics for control variables are provided from the applicable model with the greatest sample size. Refer to Table 1 for sample composition. Refer to Appendix 2 for variable definitions. We conduct univariate tests of differences in means between observations with and without an M&A KAM using t-tests (chi-square tests) for all continuous (indicator) variables. We report one-tailed p-values for signed predictions and two-tailed otherwise.

(p-values < 0.10, one-tailed). In addition, auditors issue future goodwill KAMs in the first three years after the acquisition for 50 percent of the firm-year observations in the sample. The mean for *GW_KAM_3YR* is 0.66 for the subsample of observations with acquisition-year M&A KAMs, which is significantly greater than the mean *GW_KAM_3YR* of 0.42 for observations without M&A KAMs (p-value < 0.01, one-tailed).

The mean change in operating performance from year $t-1$ to year $t+1$ (ΔOCF_{t+1}) is 12 percent, and this amount is lower for companies with (6 percent) versus without an M&A KAM (15 percent) (p-value < 0.10, one-tailed). The mean change in operating performance from year $t-1$ to year $t+2$ (ΔOCF_{t+2}) is 12 percent, and this amount is significantly lower for companies with (5 percent) versus without an M&A KAM (16 percent) (p-value = 0.05, one-tailed). The mean change in operating performance from year $t-1$ to year $t+3$ (ΔOCF_{t+3}) is 9 percent, which is significantly lower for companies with (3 percent) versus without an M&A KAM (11 percent) (p-value < 0.05, one-tailed). Fourteen percent of CEOs were forced out within three years following an M&A transaction (*CEO_TURN*), and this amount does not differ significantly between companies with versus without an M&A KAM.

Table 3 includes Pearson pairwise correlations for variables of interest from all models. With regard to our KAM disclosure model, *M&A_KAM* is positively correlated at the 0.05 level with *FOREIGN_TARGET*, but not with *RELATIVE_SIZE*, *CROSS_INDUSTRY*, or *GOODWILL*. Concerning variables from our goodwill impairment, post-acquisition performance, and CEO turnover models, *GW_IMPAIR*, *GW_IMPAIR_AMT*, and *GW_KAM_3YR* are positively correlated at the 0.05 level with *M&A_KAM*. In addition, *M&A_KAM* exhibits significant negative correlations with ΔOCF_{t+3} and (but not ΔOCF_{t+1} or ΔOCF_{t+2}) and a positive correlation with *CEO_TURN* at the 0.05 level.

4.2 M&A KAM disclosures

Table 4 presents multivariate tests of our first set of hypotheses, which predict that auditors are more likely to disclose M&A KAMs when clients complete acquisitions that are more material, involve targets with higher information asymmetry, and when there is greater overpayment risk. We estimate various specifications of Model [1] testing each hypothesis individually followed by a specification testing all of them jointly. With respect to the control variables, the results show that M&A KAMs are more likely for clients audited by the Big Four firms, with recent restatements, larger market capitalization, lower growth opportunities, higher return on assets, and more restructuring.

Tests of H1a reveal a positive and significant coefficient on *RELATIVE_SIZE* in column (1) (p-value < 0.01; one-tailed). Thus, we find support for the prediction that auditors are more likely to issue M&A KAMs for transactions that are more material. Consistent with H1b, we find a positive and significant coefficient on *FOREIGN_TARGET* in column (2) (p-value < 0.10; one-tailed). However, we fail to find a significant coefficient on *CROSS_INDUSTRY*. These results provide some support for the prediction that auditors are more likely to issue M&A KAMs when clients complete transactions involving higher information asymmetry. Turning to our tests

Table 3 Pairwise correlations

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
(1) <i>M&A_KAM</i>	1.00											
(2) <i>RELATIVE_SIZE</i>	0.02	1.00										
(3) <i>CROSS_INDUSTRY</i>	0.00	0.01	1.00									
(4) <i>FOREIGN_TARGET</i>	0.19	-0.02	0.09	1.00								
(5) <i>GOODWILL</i>	0.04	0.39	-0.03	-0.02	1.00							
(6) <i>GW_IMPAIR</i>	0.08	0.05	-0.05	0.12	0.04	1.00						
(7) <i>GW_IMPAIR_AMT</i>	0.06	0.07	-0.04	0.02	0.10	0.46	1.00					
(8) <i>GW_KAM_3YR</i>	0.21	-0.09	-0.07	0.12	0.05	0.23	0.10	1.00				
(9) $\Delta OCF_t + 1$	-0.06	0.18	-0.01	0.01	0.17	-0.06	-0.01	-0.06	1.00			
(10) $\Delta OCF_t + 2$	-0.03	0.14	-0.03	0.02	0.13	-0.06	-0.01	-0.06	0.89	1.00		
(11) $\Delta OCF_t + 3$	-0.09	0.23	-0.02	0.00	0.19	-0.05	0.01	-0.08	0.80	0.92	1.00	
(12) <i>CEO_TURN</i>	0.06	-0.03	-0.01	0.04	0.05	0.11	0.04	0.14	-0.05	-0.03	-0.06	1.00

This table presents Pearson pairwise correlations for variables of interest from all models. Refer to Appendix 2 for variable definitions. Bold text indicates significance at the 5% level or better.

Table 4 M&A KAM disclosures

Dependent Variable:	(1)		(2)		(3)		(4)	
	Hypothesis 1a	t-stat	Hypothesis 1b	t-stat	Hypothesis 1c	t-stat	Joint test	t-stat
M&A_KAM	Coef		Coef		Coef		Coef	
Independent Variables:								
<i>RELATIVE_SIZE</i>							0.045***	2.67
<i>CROSS_INDUSTRY</i>	0.057***	3.57	0.017	1.08			0.016	1.01
<i>FOREIGN_TARGET</i>			0.034*	1.64			0.030*	1.43
<i>GOODWILL</i>					0.188***	3.56	0.119**	2.06
Control Variables:								
<i>BIG4</i>	0.070*	1.83	0.069*	1.80	0.068*	1.77	0.066*	1.72
<i>ΔAUDITOR</i>	-0.048	-1.06	-0.039	-0.88	-0.044	-0.98	-0.044	-0.98
<i>RESTATE</i>	0.100	1.51	0.095	1.43	0.093	1.40	0.107*	1.66
<i>ASSETS</i>	0.052***	4.49	0.040***	3.49	0.050***	4.28	0.050***	4.21
<i>BTM</i>	-0.090***	-2.61	-0.076**	-2.23	-0.060*	-1.71	-0.069*	-1.93
<i>SALES_GROWTH</i>	0.006	0.40	0.001	0.07	0.005	0.31	0.007	0.45
<i>FOREIGN_SALES</i>	0.085	1.62	0.057	1.04	0.087*	1.65	0.060	1.08
<i>LEVERAGE</i>	-0.121	-1.08	-0.102	-0.92	-0.073	-0.64	-0.117	-1.06
<i>ROA</i>	0.154	1.39	0.045	0.44	0.126	1.13	0.197*	1.73
<i>RESTRUCTURING</i>	2.251*	1.68	2.211*	1.68	1.963	1.47	2.011	1.55
Intercept	0.016	0.10	0.069	0.40	-0.021	-0.13	-0.041	-0.26
Fixed effects	Industry and year		Industry and year		Industry and year		Industry and year	
Adjusted R ²	0.16		0.15		0.16		0.17	
N	1025		1025		1025		1025	

This table presents results of testing H1a, H1b, and H1c and reports the results of estimating a linear probability model in which the dependent variable is *M&A_KAM*, an indicator variable equal to one if the audit report includes an M&A KAM. Column (1), (2), and (3) tests hypotheses 1a, 1b, and 1c individually, while column 4 conducts a joint test of hypotheses 1a–1c. Refer to Appendix 2 for all other variable definitions. Results are estimated with standard errors clustered at the company level. *, **, and *** indicate statistical significance at the 0.10, 0.05, or 0.01 level, respectively. Significance is based on one-tailed p-values for signed predictions and two-tailed otherwise.

of H1c, we find that the coefficient on *GOODWILL* is positive and significant in column (3) (p-value < 0.01; one-tailed). In terms of economic significance, this result translates to a 5 percent increase in the likelihood of auditors issuing an M&A KAM for each one standard deviation increase in *GOODWILL* (untabulated). In column (4), we conduct a joint test of H1a, H1b, and H1c, simultaneously including *RELATIVE_SIZE*, *CROSS_INDUSTRY*, *FOREIGN_TARGET*, and *GOODWILL* in Model [1]. We continue to find positive and significant coefficients on each variable (p-values < 0.10 or better), except for *CROSS_INDUSTRY*. Overall, these results are consistent with the prediction that auditors are more likely to make M&A KAM disclosures when acquisitions are more material, involve targets with higher information asymmetry, and in acquisitions with a greater risk of overpayment.¹⁶

4.3 Goodwill impairment

Table 5 presents multivariate tests of H2a, testing the relation between acquisition-year M&A KAMs and future goodwill impairment. After controlling for variables capturing materiality, information asymmetry, and overpayment risk related to the M&A transaction, as well as investors' market reaction to the M&A transaction announcement and other determinants of acquisition failure, the results show some support for H2a. The results in column (1) show that the coefficient on *M&A_KAM* is not significantly different from zero. This suggests that M&A KAMs do not predict a significantly higher probability of goodwill impairment. However, the results in columns (2) and (3) show that the coefficients on *M&A_KAM* are positive and significant (p-value < 0.05 and 0.01, respectively). The results in column (2) suggest that, all else equal, acquirers with acquisition-year M&A KAMs report significantly larger goodwill impairment losses. Turning to column (3), the results suggest that auditors are significantly more likely to issue future goodwill KAMs after making M&A-related KAM disclosures in the acquisition year, consistent with a higher risk of future goodwill impairment (Andreicovici et al. 2025).

4.4 Post-acquisition performance

Table 6 presents multivariate tests of H2b, with columns (1), (2), and (3) providing results of changes in post-acquisition operating performance in years $t+1$, $t+2$, and $t+3$, respectively. The adjusted R^2 is between 0.09 and 0.12, and we find positive coefficients on *GOODWILL* across all model specifications (p-values < 0.05; one-tailed), which is consistent with post-acquisition realization of anticipated synergies and other benefits from the acquisition. Consistent with H2b, we

¹⁶ The magnitude and significance of the coefficients on *RELATIVE_SIZE* and *GOODWILL* decline in column (4) when we include them in the model together. Although these variables are positively correlated, untabulated analysis of variance inflation factors suggest little influence of multicollinearity (all VIFs are less than 2.0).

Table 5 M&A KAMs and future goodwill impairments

		(1)		(2)		(3)	
Dependent Variables:		<i>GW_IMPAIR</i>		<i>GW_IMPAIR_AMT</i>		<i>GW_KAM_3YR</i>	
	Predictions	Coef	t-stat	Coef	t-stat	Coef	t-stat
Independent Variable:							
<i>M&A_KAM</i>	H2a [+]	-0.001	-0.02	0.018**	1.78	0.129***	3.53
Control Variables:							
<i>RELATIVE_SIZE</i>	?	0.024	0.61	-0.015	-1.57	-0.038	-1.11
<i>CROSS_INDUSTRY</i>	?	-0.015	-0.75	-0.005	-1.09	0.012	0.67
<i>FOREIGN_TARGET</i>	?	0.033	1.43	0.002	0.52	0.043*	1.96
<i>GOODWILL</i>	+	0.199**	2.03	0.092**	2.16	0.149*	1.55
<i>AB_ANN_RET</i>	-	-0.345**	-1.81	-0.062	-1.01	-0.153	-0.69
<i>ASSETS</i>	?	0.020	1.62	-0.001	-0.35	-0.017	-1.44
<i>BTM</i>	?	0.010	0.21	0.027	1.47	-0.010	-0.21
<i>LEVERAGE</i>	?	-0.010	-0.08	0.007	0.24	-0.255**	-2.00
<i>ALL_STOCK</i>	-	-0.156**	-2.26	0.021	0.59	-0.071	-0.69
<i>ALL_CASH</i>	-	-0.047	-1.03	0.004	0.27	-0.094**	-2.29
<i>#ACQUISITIONS</i>	?	0.014	0.74	0.005	1.06	-0.013	-0.84
<i>#OTHER_KAMS</i>	?	-0.012	-1.15	-0.001	-0.25	0.000	0.01
<i>GW_KAM</i>	+	0.116**	2.32	0.002	0.15	0.344***	7.43
Intercept		0.006	0.04	-0.060*	-1.94	0.212	1.06
Fixed effects		Industry and year		Industry and year		Industry and year	
Adjusted R ²		0.04		0.03		0.24	
N		896		896		896	

This table presents results of testing H2a and presents the results of estimating a linear probability model in column (1) in which the dependent variable is *GW_IMPAIR*, an indicator variable equal to one if the company records a goodwill impairment within the three years following an M&A transaction; zero otherwise. In column (2), we estimate an OLS regression model in which the dependent variable is *GW_IMPAIR_AMT*, which equals the total amount of goodwill impairment divided by sales in years $t+1$, $t+2$, and $t+3$. In column (3), we estimate a linear probability model in which the dependent variable is *GW_KAM_3YR*, an indicator variable equal to one if the audit report includes an auditor KAM disclosure relating to goodwill within the three years following an M&A transaction; zero otherwise. Refer to Appendix 2 for all other variable definitions. Results are estimated with standard errors clustered at the company level. *, **, and *** indicate statistical significance at the 0.10, 0.05, or 0.01 level, respectively. Significance is based on one-tailed p-values for signed predictions and two-tailed otherwise.

find that the coefficients on the *M&A_KAM* * *GOODWILL* interaction terms are negative across years $t+1$, $t+2$, and $t+3$ (p-values < 0.01; one-tailed). These results imply that, when the auditor issues an M&A KAM in the acquisition year, the association between the acquired goodwill and changes in the company's operating performance is significantly lower in the three subsequent years as compared to when the auditor does not issue an M&A KAM. In other words, the findings are consistent with uncertainty expressed in the M&A KAM disclosures predicting a higher likelihood that acquirers will fail to successfully execute their acquisition strategies. Interpreting the economic effects of these results at the mean amount of acquired goodwill in our sample (0.11), the *M&A_KAM* * *GOODWILL* interaction translates to an economically significant reduction in operating cash flows ranging

Table 6 M&A KAMs and future operating cash flow performance

Dependent Variables:		(1)		(2)		(3)	
		ΔOCF_{t+1}		ΔOCF_{t+2}		ΔOCF_{t+3}	
	Predictions	Coef	t-stat	Coef	t-stat	Coef	t-stat
Independent Variables:							
<i>M&A_KAM</i>	?	0.046	0.65	0.042	0.57	0.016	0.34
<i>GOODWILL</i>	+	0.769**	2.23	1.037***	2.40	0.560**	2.01
<i>M&A_KAM * GOODWILL</i>	H2b [-]	-0.100***	-2.48	-1.196***	-2.54	-0.763***	-2.53
Control Variables:							
<i>RELATIVE_SIZE</i>	?	0.081	1.00	0.059	0.57	0.089	1.24
<i>CROSS_INDUSTRY</i>	?	0.010	0.32	0.017	0.57	0.002	0.07
<i>FOREIGN_TARGET</i>	?	0.073**	2.18	0.084**	2.33	0.065**	2.59
<i>AB_ANN_RET</i>	+	-1.061*	-1.82	-0.688	-1.01	-0.715*	-1.72
<i>ASSETS</i>	?	-0.086***	-3.10	-0.084***	-2.81	-0.066***	-3.16
<i>BTM</i>	?	-0.133	-1.31	-0.030	-0.27	-0.051	-0.70
<i>LEVERAGE</i>	?	-0.009	-0.03	0.069	0.23	0.296	1.32
<i>ALL_STOCK</i>	+	0.057	0.32	0.118	0.59	0.318**	1.76
<i>ALL_CASH</i>	+	0.161**	1.66	0.115**	1.86	0.051	1.21
<i>#ACQUISITIONS</i>	?	-0.014	-0.59	-0.016	-0.65	-0.009	-0.49
<i>#OTHER_KAMS</i>	?	-0.004	-0.27	0.000	0.02	0.001	0.14
<i>GW_KAM</i>	?	-0.009	-0.25	-0.002	-0.04	0.038	1.01
Intercept		1.702	1.36	1.755	1.31	1.131	1.35
Fixed effects		Industry and year		Industry and year		Industry and year	
Adjusted R ²		0.10		0.09		0.12	
N		953		919		815	

This table presents results of testing H2b and reports the results of estimating OLS regression models in which the dependent variable is ΔOCF_{t+1} , ΔOCF_{t+2} , and ΔOCF_{t+3} , in columns (1), (2), and (3), respectively. ΔOCF is calculated as the change in operating performance from year $t-1$ to year $t+n$, where operating performance equals operating cash flows divided by sales. Refer to Appendix 2 for all other variable definitions. Results are estimated with standard errors clustered at the company level. *, **, and *** indicate statistical significance at the 0.10, 0.05, or 0.01 level, respectively. Significance is based on one-tailed p-values for signed predictions and two-tailed otherwise.

from 8.4 to 13.2 percent of sales when the auditor issues an M&A KAM as compared to when the auditor does not (untabulated).

Importantly, our multivariate tests reveal that the association between M&A KAM disclosures and poor post-acquisition performance is incremental to controlling for variables that proxy for the materiality, information asymmetry, and overpayment risk related to the M&A transaction, as well as investors' market reaction to the M&A transaction announcement. This suggests that auditors' M&A KAM

disclosures contain incremental information about future post-acquisition performance that investors' initial expectations do not reflect.¹⁷

4.5 Forced CEO turnover

Table 7 Panel A presents multivariate results for testing H3, which predicts that the association between poor post-acquisition performance and CEO turnover will be stronger (i.e., more positive) when the auditor issues an M&A KAM in the year of acquisition. Consistent with H3, we find positive and significant coefficients on $M\&A_KAM * \Delta OCF_NEGATIVE$ in column (1) and $M\&A_KAM * GW_IMPAIR_AMT$ in column (2) (p-values < 0.05, one-tailed). This implies that the likelihood of forced CEO turnover is incrementally higher when there is poor post-acquisition performance and the auditor issues an M&A KAM in the acquisition year (compared to when the auditor does not). Speaking in terms of economic significance, the results in column (1) translate to a mean 10 percent greater likelihood of forced CEO turnover when the company experiences poor post-acquisition performance and the auditor issues an M&A KAM in the year of acquisition as compared to when the auditor does not (untabulated). Importantly, the association between M&A KAM disclosures and forced CEO turnover is *incremental* to controlling for investors' market reaction to the M&A transaction announcement. Thus, these findings suggest that auditor M&A KAM disclosures contain information related to failed M&A beyond what investors' initial expectations reflect.

We perform a falsification test of H3 by replacing the dependent variable in Model [3] with *UNFORCED*, an indicator variable equal to one if the CEO left the company within the three years following an M&A transaction and did *not* meet the criteria for classification as a forced turnover and zero otherwise. We do not expect an association between M&A KAM disclosures, interactions with poor post-acquisition performance, and unforced CEO turnover. Table 7 Panel B presents the multivariate results. Consistent with our expectation, we find the coefficients on $M\&A_KAM * \Delta OCF_NEGATIVE$ and $M\&A_KAM * GW_IMPAIR_AMT$ are not significantly different from zero. This further supports the validity of the results for Model [3] as well as the method to identify forced versus unforced CEO turnover from Parrino (1997) and Peters and Wagner (2014).

¹⁷ Lennox et al. (2023) posit that KAM disclosures are not associated with stock price reactions because investors impound information reflected in those disclosures before managers release financial statements. It is possible that the initial stock price reaction at the M&A announcement date includes investors' expectations that the auditor will issue an M&A KAM for the acquisition-year financial statements. We do not test whether M&A KAM disclosures produce incremental price reactions, but our results suggest that investors' initial assessment of uncertainty later reflected in KAMs is incomplete.

Table 7 M&A KAMs and future CEO turnover

Panel A: Forced CEO turnover					
Dependent Variable:		(1) Decrease in operating performance		(2) Goodwill impairment	
<i>CEO_TURN</i>	Predictions	Coef	t-stat	Coef	t-stat
Independent Variables:					
<i>ΔOCF_NEGATIVE</i>	+	−0.008	−0.21		
<i>GW_IMPAIR_AMT</i>	+			0.021	0.23
<i>M&A_KAM</i>	?	−0.072*	−1.78	−0.037	−1.01
<i>M&A_KAM * ΔOCF_NEGATIVE</i>	H3 [+]	0.103**	1.70		
<i>M&A_KAM * GW_IMPAIR_AMT</i>	H3 [+]			0.561**	1.82
Control Variables:					
<i>RELATIVE_SIZE</i>	?	0.019	0.68	0.020	0.68
<i>CROSS_INDUSTRY</i>	?	0.031	1.45	0.033	1.58
<i>FOREIGN_TARGET</i>	?	−0.003	−0.12	−0.003	−0.13
<i>GOODWILL</i>	+	−0.019	−0.24	−0.027	−0.33
<i>AB_ANN_RET</i>	−	−0.201	−0.98	−0.183	−0.89
<i>ROA</i>	+	0.412***	2.85	0.413***	2.82
<i>ΔROA_{t-1}</i>	+	−0.211	−1.01	−0.226	−1.06
<i>ΔSALES_{t-1}</i>	+	0.149**	2.01	0.155**	2.06
<i>ΔRET_{t-1}</i>	+	0.006	0.45	0.005	0.37
<i>ASSETS</i>	?	0.007	0.61	0.007	0.61
<i>BTM</i>	?	0.018	0.41	0.014	0.31
<i>LEVERAGE</i>	?	0.139	1.36	0.130	1.28
<i>LOSS</i>	+	0.083*	1.36	0.081*	1.34
<i>RESTATE</i>	+	−0.080	−1.44	−0.084	−1.40
<i>#ACQUISITIONS</i>	?	−0.008	−0.38	−0.009	−0.47
<i>#OTHER_KAMS</i>	?	0.004	0.40	0.003	0.31
<i>GW_KAM</i>	+	0.009	0.21	0.005	0.11
<i>CEO_TENURE</i>	−	−0.007***	−2.73	−0.007***	−2.71
<i>CEO_CHAIR</i>	?	−0.067	−0.88	−0.070	−0.93
Intercept		−0.087	−0.85	−0.068	−0.77
Fixed effects		Industry and year		Industry and year	
Adjusted R ²		0.05		0.05	
N		580		580	

Table 7 (Continued)

Panel B: Falsification test – unforced CEO turnover

Dependent Variables:		(1)		(2)	
		Decrease in operating performance		Goodwill impairment	
<i>UNFORCED</i>	Predictions	Coef	t-stat	Coef	t-stat
Independent Variables:					
<i>ΔOCF_NEGATIVE</i>	+	0.025	0.65		
<i>GW_IMPAIR_AMT</i>	+			0.075	0.59
<i>M&A_KAM</i>	?	0.088	1.51	0.077*	1.76
<i>M&A_KAM * ΔOCF_NEGATIVE</i>	H3 [+]	−0.015	−0.23		
<i>M&A_KAM * GW_IMPAIR_AMT</i>	H3 [+]			0.094	0.34
Control Variables:					
<i>RELATIVE_SIZE</i>	?	0.026	0.95	0.027	0.98
<i>CROSS_INDUSTRY</i>	?	0.014	0.70	0.015	0.76
<i>FOREIGN_TARGET</i>	?	0.036	1.53	0.035	1.48
<i>GOODWILL</i>	+	−0.049	−0.47	−0.057	−0.54
<i>AB_ANN_RET</i>	−	−0.024	−0.10	−0.007	−0.03
<i>ROA</i>	+	0.115	0.75	0.125	0.81
<i>ΔROAt-1</i>	+	0.002	0.01	−0.001	−0.01
<i>ΔSALES_{t-1}</i>	+	−0.127	−1.28	−0.122	−1.21
<i>ΔRET_{t-1}</i>	+	0.004	0.29	0.003	0.21
<i>ASSETS</i>	?	−0.005	−0.48	−0.005	−0.45
<i>BTM</i>	?	−0.036	−0.96	−0.040	−1.06
<i>LEVERAGE</i>	?	0.023	0.19	0.015	0.13
<i>LOSS</i>	+	−0.022	−0.38	−0.024	−0.42
<i>RESTATE</i>	+	0.107	1.22	0.105	1.21
<i>#ACQUISITIONS</i>	?	0.003	0.17	0.002	0.13
<i>#OTHER_KAMS</i>	?	−0.004	−0.45	−0.004	−0.41
<i>GW_KAM</i>	+	0.004	0.10	0.003	0.06
<i>CEO_TENURE</i>	−	0.014***	3.24	0.014***	3.23
<i>CEO_CHAIR</i>	?	0.059	0.44	0.062	0.47
Intercept		0.278	1.03	0.292	1.06
Fixed effects		Industry and year		Industry and year	
Adjusted R ²		0.05		0.05	
N		580		580	

This table presents results of testing H3 and reports the results of estimating linear probability models in which the dependent variable is *CEO_TURN* and *UNFORCED* in Panels A and B, respectively. *CEO_TURN* is an indicator variable equal to one if the CEO was forced out within the three years following an M&A transaction. *UNFORCED* is an indicator variable equal to one if the CEO left the company within the three years following an M&A transaction and did not meet the criteria for classification as a forced turnover. Refer to Appendix 2 for all other variable definitions. Results are estimated with standard errors clustered at the company level. *, **, and *** indicate statistical significance at the 0.10, 0.05, or 0.01 level, respectively. Significance is based on one-tailed p-values for signed predictions and two-tailed otherwise.

4.6 Robustness tests

Our current sample includes all LSE-listed companies for the full sample period of September 2013 through December 2020, as early adoption of expanded auditor reporting requirements by non-main market companies was permitted. We perform robustness tests to rule out the potential alternative explanation that our findings are driven by the effects of voluntary adoption by removing AIM companies from our sample during their voluntary adoption period (i.e., September 2013–June 2017). Therefore, the sample for these tests includes main-market companies completing acquisitions between September 2013 and December 2020 and AIM companies completing acquisitions after June 2017 through December 2020.

Table 8 presents results of our models using this alternative sample of mandatory adopters. Panel A includes the results of our M&A KAM disclosure model. Consistent with H1a, we find a positive and significant coefficient on *RELATIVE_SIZE* in column (1) (p-value < 0.05; one-tailed). Turning to our tests of H1b, we find a positive and significant coefficient on *FOREIGN_TARGET* in column (2) (p-value < 0.10; one-tailed). However, we fail to find a significant coefficient on *CROSS_INDUSTRY*. Finally, consistent with H1c, we find that the coefficient on *GOODWILL* is positive and significant in column (3) (p-value < 0.01; one-tailed). In column (4), we conduct a joint test of H1a, H1b, and H1c, simultaneously and continue to find positive coefficients on each variable, though some lose their significance, potentially due to the decreased sample size. Overall, these results are consistent with the prediction that auditors are more likely to make M&A KAM disclosures when acquisitions are more material, involve targets with higher information asymmetry, and in acquisitions with a greater risk of overpayment.

Table 8 Panel B includes the results of our goodwill impairment model. The results in column (1) show that the coefficient on *M&A_KAM* is not significantly different from zero, suggesting M&A KAMs do not predict a significantly higher probability of goodwill impairment. However, the results in columns (2) and (3) show that the coefficients on *M&A_KAM* are positive and significant (p-value < 0.05 and 0.01, respectively), suggesting acquirers with acquisition-year M&A KAMs report significantly larger goodwill impairment losses and are significantly more likely to issue future goodwill KAMs, consistent with H2a. Panel C includes the results of our post-acquisition performance model. Consistent with H2b, we find that the coefficients on the *M&A_KAM* * *GOODWILL* interaction terms are negative across years $t+1$, $t+2$, and $t+3$ (p-values < 0.01; one-tailed). These results imply that, when the auditor issues an M&A KAM in the acquisition year, the association between the acquired goodwill and changes in the company's operating performance is significantly lower in the three subsequent years as compared to when the auditor does not.

Table 8 Panel D presents the results of our forced CEO turnover model. Consistent with H3, we find a positive and significant coefficient on *M&A_KAM* * $\Delta OCF_NEGATIVE$ in column (1) (p-value < 0.10, one-tailed). This implies that the likelihood of forced CEO turnover is incrementally higher when there is poor post-acquisition performance and the auditor issues an M&A KAM in the acquisition year (compared to when the auditor does not). Overall, our inferences remain

Table 8 Robustness tests

Panel A: M&A KAM disclosures									
Dependent Variable: <i>M&A_KAM</i>	Predictions	(1)		(2)		(3)		(4)	
		Hypothesis 1a Coef	t-stat	Hypothesis 1b Coef	t-stat	Hypothesis 1c Coef	t-stat	Joint test Coef	t-stat
Independent Variables:									
<i>RELATIVE_SIZE</i>	H1a [+]	0.026**	1.93					0.016	1.14
<i>CROSS_INDUSTRY</i>	H1b [+]			0.013	0.83			0.014	0.90
<i>FOREIGN_TARGET</i>	H1b [+]			0.032*	1.32			0.025	1.04
<i>GOODWILL</i>	H1c [+]					0.491***	3.67	0.444***	3.31
Controls		Yes		Yes		Yes		Yes	
Fixed effects		Industry and year		Industry and year		Industry and year		Industry and year	
Adjusted R ²		0.07		0.07		0.09		0.09	
N		726		726		726		726	
Panel B: M&A KAMs and future goodwill impairments									
Dependent Variable: <i>M&A_KAM</i>	Predictions	(1)		(2)		(3)			
		<i>GW_IMPAIR</i> Coef	t-stat	<i>GW_IMPAIR_AMT</i> Coef	t-stat	<i>GW_KAM_3YR</i> Coef	t-stat		
Independent Variables:									
<i>M&A_KAM</i>	H2a [+]	0.014	0.36	0.014**	1.76	0.122***	3.16		
Controls		Yes		Yes		Yes			
Fixed effects		Industry and year		Industry and year		Industry and year			
Adjusted R ²		0.07		0.03		0.37			
N		634		634		634			

Table 8 (continued)

Panel C: M&A KAMs and future operating cash flow performance					
Dependent Variables:	Predictions	(1)		(2)	
		ΔOCF_{t+1}	t-stat	ΔOCF_{t+2}	ΔOCF_{t+3}
Independent Variables:					
<i>M&A_KAM</i>	?	0.070	0.79	0.059	0.074
<i>GOODWILL</i>	+	2.023**	2.20	2.310**	2.677***
<i>M&A_KAM * GOODWILL</i>	H2b [-]	-2.373***	-2.58	-2.566***	-2.939***
Controls		Yes		Yes	Yes
Fixed effects		Industry and year		Industry and year	Industry and year
Adjusted R ²		0.16		0.13	0.26
N		682		663	573
Panel D: M&A KAMs and future forced CEO turnover					
Dependent Variable:	Predictions	(1)		(2)	
		Decrease in operating performance	t-stat	Goodwill impairment	t-stat
Independent Variables:					
<i>CEO_TURN</i>					
$\Delta OCF_NEGATIVE$	+	-0.033	-0.60		
<i>GW_IMPAIR_AMT</i>	+			0.264	0.58
<i>M&A_KAM</i>	?	-0.077*	-1.67	-0.032	-0.81
<i>M&A_KAM * ΔOCF_NEGATIVE</i>	H3 [+]	0.098*	1.42		
<i>M&A_KAM * GW_IMPAIR_AMT</i>	H3 [+]			0.152	0.28

Table 8 (continued)

Controls	Yes	Yes
Fixed effects	Industry and year	Industry and year
Adjusted R ²	0.04	0.04
N	404	404

This table presents results of robustness tests re-estimating the same models within Tables 4, 5, 6 and 7, included in Panels A–D, respectively, for an alternative sample that excludes voluntary adopters. This sample includes Main Market companies from September 2013–December 2020 and Alternative Investment Market Companies from June 2017–December 2020, thereby excluding Alternative Investment Companies from September 2013–June 2017 during the voluntary adoption period. Refer to Appendix 2 for variable definitions. Results are estimated with standard errors clustered at the company level. *, **, and *** indicate statistical significance at the 0.10, 0.05, or 0.01 level, respectively. Significance is based on one-tailed p-values for signed predictions and two-tailed otherwise.

consistent, although we fail to find a significant coefficient on $M\&A_KAM * GW_IMPAIR_AMT$ in column (2), potentially due to the decreased sample size. In sum, our findings do not appear to be driven by voluntary adoption effects.

5 Conclusion

We examine whether auditor KAM disclosures communicate information around the uncertainty implicit in M&A transactions and investigate how these disclosures predict the success or failure of managers' acquisition strategies. We find that auditors make more M&A KAM disclosures when acquirers complete transactions that (1) have a more material impact on the financial statements, (2) involve targets with greater levels of information asymmetry, and (3) are subject to greater overpayment risk. These results reveal that auditors' M&A KAM disclosures reflect variation in the level of uncertainty underlying M&A transactions.

When there is greater economic uncertainty about the future payoff to a CEO's M&A investment decision, there is a wider range of future possible outcomes (both good and bad). Consistent with this, we find that M&A KAM disclosures predict poor post-acquisition performance and forced CEO turnover. When auditors issue an M&A KAM in the acquisition year, acquirers record larger future goodwill impairments, and the association between acquired goodwill and changes in companies' subsequent operating performance is significantly lower compared to when auditors do not issue an M&A KAM. These findings suggest that M&A KAMs reflect uncertainty about the realization of expected benefits from the acquisition. In addition, the likelihood of forced CEO turnover is incrementally higher when subsequent operating performance is poor and auditors issue an M&A KAM compared to when they do not.

Our study makes several contributions to the literature. First, we provide new insights on the role of auditors in M&A transactions by showing that auditor KAM disclosures highlight sources of uncertainty surrounding these transactions. Second, we contribute to research identifying factors that predict success versus failure of M&A transactions. We also contribute to the literature examining CEO turnover following failed acquisitions by providing the first evidence linking disclosures highlighting uncertainty to post-acquisition forced CEO turnover. In addition, we provide incremental contributions to the KAM literature. The results reveal that the information content of KAMs reflects the level of uncertainty around M&A transactions, consistent with regulators' perspective that moving away from the traditional audit report will provide useful information to investors. We also document that M&A KAMs provide an early warning' sign indicating the probability of failure of M&A transactions, along with corporate governance responses in the form of forced CEO turnover. Our paper differs from others in that we narrow our focus to the topical area of M&A and develop a KAM disclosure model that focuses on the characteristics of the M&A transaction itself as opposed to only the characteristics of the auditor and the company considered in prior literature.

This study is also subject to certain limitations. Our data derive from the United Kingdom, which provides us a longer time series of data than is available in the United States, where CAM reporting has more recently begun. The U.S. setting features different regulatory requirements and legal exposure for auditors, which likely

account for variation in the number, the types, and the level of detail in KAMs versus CAMs (Nylen et al. 2024). It is beyond the scope of our study to test whether and how the results might generalize to other jurisdictions. Despite this, we exploit the features of M&A as an empirical setting where uncertainty and accounting complexity are high. This allows us to conduct powerful tests of M&A KAM disclosures, answering the call in prior research to examine whether KAMs have unique effects on complex aspects of financial reporting (Minutti-Meza 2021). We encourage future research to explore effects of KAMs in other regulatory and legal settings.

Appendix 1. Excerpts from M&A KAM disclosures

The Daily Mail's FYE 2015 audit report

KAM Title: Accounting for acquisitions and disposals.

Description:

Refer to the Audit Committee report on pages 54 to 60 and to Notes 17 and 18 in the Consolidated Financial Statements. The Group continues to make significant investment in business acquisitions. During the year the Group made a number of acquisitions including Elite Daily Inc, Petrotranz Inc, Starfish Retention Solutions Inc, Locus Energy Inc, and TreppPort Inc.

IFRS 3 'Business combinations' requires the fair value of consideration paid to be allocated firstly to the fair value of net assets acquired, then to the fair value of separately identifiable intangible assets and finally to goodwill, which involves a number of estimates and judgements. The Group's acquisitions also include earn-out clauses and future rights or obligations to purchase an additional stake which may impact accounting judgments on initial recognition...

We focused our work on the acquisitions and disposals during the year with the largest value of purchase consideration and those where the terms were not straightforward...

Response:

We read the minutes of the Investment and Finance Committee and inquired of the directors to identify acquisition and disposals during the year. For all material and complex transactions we obtained and read the sale and purchase agreements ("SPAs"). In testing acquisitions during the year we:

- assessed the appropriateness of the directors' identification of intangible assets acquired by reviewing the SPAs to identify any assets listed, forming an independent expectation of the assets based on those in comparative companies and comparing the ratio of the value of goodwill acquired to the intangible assets to comparable transactions;
- obtained the directors' calculation of the fair value of intangible assets acquired, and corroborated the appropriateness of the assumptions over royalty rates, levels of customer attrition, useful economic lives and forecast cash flows to external benchmarks, historical performance and the acquisition plans; and
- verified the fair value of consideration paid of acquisitions, including any deferred or contingent element, to cash transactions and the SPAs.

RWS Holding's FYE 2017 audit report

KAM Title: Acquisition accounting for LUZ.¹⁸

Description:

Refer to page 24 (audit committee), note 22 of the financial statements and note 1 for the Directors' disclosures of the related accounting policies, judgements and estimates for further information.

The Group acquired LUZ Inc. in February 2017 for a cash consideration of US\$82.5 million. Accounting for the acquisition required a fair value exercise to assess the assets and liabilities acquired, including valuing any separately identifiable intangible assets with the residual balance recognized as goodwill. The valuation of identified intangibles can be a subjective process and as such was an area of focus for us.

Fair value of identified intangibles

Management identified US\$35.9 million of intangible assets in respect of LUZ's customer relationships, software, noncompete agreements and order backlog.

The fair value of these intangible assets was judgmental as it used valuation techniques that require management assumptions including customer attrition rates, growth rates for existing customer revenues, forecast profitability levels and an appropriate discount rate.

Response:

To address this risk we:

- a. Examined the acquisition agreements and other documents including due diligence reports;
- b. Ensured accounting is in accordance with IFRS 3 Business Combinations;
- c. Performed testing procedures on the acquired opening balance sheet including management adjustments;
- d. Tested the fair value adjustments, working with both management and management's expert to verify and challenge key assumptions; and
- e. Utilized our own expert to support the audit of fair value adjustments.

Fair value of identified intangibles

To address this risk we:

- a. Assessed the completeness and quantum of intangible assets identified by management against our own expectations, formed from review of the due diligence reports prepared by management's professional advisors during the acquisition, and disclosures surrounding the rationale for the transactions.

¹⁸ LUZ Inc. is a U.S. company and the cash purchase consideration in this transaction is denominated in U.S. dollars.

- b. Assessed the work performed on the purchase price allocation by utilizing our in-house specialists to evaluate management's valuation of the identified assets. Specifically, we reviewed the methodology adopted, compared the assumptions made on attrition and recoverability with historical patterns in the business to verify that assumptions were reasonable, considered the discount rate used and verified the mathematical accuracy of the calculations.
- c. Corroborated the value of intangibles by performing an overall sense-check of the level of residual goodwill arising on the transaction, by considering the level of resulting goodwill as a proportion of the total consideration paid as compared to similar transactions in the market.

Appendix 2. Variable definitions

Variable Name	Definition
<i>Dependent variables are in bold by model number</i>	
Model [1]:	
M&A_KAM	Indicator variable equal to one if the audit report includes an auditor KAM disclosure relating to accounting for business combinations in the acquisition completion year; zero otherwise
<i>RELATIVE_SIZE</i>	The purchase price scaled by the acquiring company's market value of equity at the end of year $t - 1$, aggregated by company-year
<i>CROSS_INDUSTRY</i>	The number of cross-industry deals (i.e., when the two-digit SIC codes of the target and acquiring company differ) completed for each company-year
<i>FOREIGN_TARGET</i>	The number of foreign target deals (i.e., when the target company's nation is outside the United Kingdom) completed for each company-year
<i>GOODWILL</i>	The change in goodwill from year $t - 1$ to year t (plus impairment in year t) scaled by the acquiring company's total assets at the end of year $t - 1$
<i>BIG4</i>	Indicator variable equal to one if PwC, EY, Deloitte, or KPMG performed the audit; zero otherwise
<i>ΔAUDITOR</i>	Indicator variable equal to one if the auditor changed in the current year; zero otherwise
<i>RESTATE</i>	Indicator variable equal to one if the issuer disclosed a restatement in the current or prior two fiscal years; zero otherwise
<i>ASSETS</i>	The natural log of total assets for the acquiring company at the end of year $t - 1$
<i>BTM</i>	The book value of equity divided by the market value of equity of the acquiring company at the end of year $t - 1$
<i>SALES_GROWTH</i>	Change in sales of the acquiring company in year $t - 1$ divided by sales for the acquiring company in year $t - 1$
<i>FOREIGN_SALES</i>	Percentage of foreign sales calculated as international sales divided by total geographic segment sales for the acquiring company in year $t - 1$

Variable Name	Definition
<i>LEVERAGE</i>	The ratio of total debt to total assets of the acquiring company at the end of year $t - 1$
<i>ROA</i>	Net income divided by average total assets of the acquiring company in year $t - 1$
<i>RESTRUCTURING</i>	Restructuring expense divided by total assets of the acquiring company in year $t - 1$
Model [2a]:	
<i>GW_IMPAIR</i>	Indicator variable equal to one if the company records a goodwill impairment within the three years following an M&A transaction; zero otherwise
<i>GW_IMPAIR_AMT</i>	The total amount of goodwill impairment divided by sales in years $t + 1$, $t + 2$, and $t + 3$
<i>GW_KAM_3YR</i>	Indicator variable equal to one if the audit report includes an auditor KAM disclosure relating to goodwill within the three years following an M&A transaction; zero otherwise
Model [2b]:	
ΔOCF_{t+1}	Change in operating performance from year $t - 1$ to year $t + 1$, where operating performance equals operating cash flows divided by sales
ΔOCF_{t+2}	Change in operating performance from year $t - 1$ to year $t + 2$, where operating performance equals operating cash flows divided by sales
ΔOCF_{t+3}	Change in operating performance from year $t - 1$ to year $t + 3$, where operating performance equals operating cash flows divided by sales
<i>AB_ANN_RET</i>	The sum of the three-day abnormal returns in the period surrounding the M&A announcement date, calculated each day as the company returns minus same-day returns for the London Stock Exchange (LSE) value-weighted portfolio. When a company has multiple acquisitions in the same fiscal year, we calculate the variable as the sum of abnormal returns across acquisitions for the fiscal year
<i>ALL_STOCK</i>	Indicator variable equal to one when the consideration consists solely of stock; zero otherwise. When a company has multiple acquisitions in the same fiscal year, we calculate the variable as a weighted average by transaction size
<i>ALL_CASH</i>	Indicator variable equal to one when the consideration consists solely of cash; zero otherwise. When a company has multiple acquisitions in the same fiscal year, we calculate the variable as a weighted average by transaction size
<i>#ACQUISITIONS</i>	The total number of completed M&A transactions by the acquirer in year t
<i>GW_KAM</i>	Indicator variable equal to one if the audit report includes an auditor KAM disclosure relating to goodwill in year $t - 1$; zero otherwise
<i>#OTHER_KAMS</i>	The total number of KAMs covering topics other than business combinations or goodwill included in the audit report in the acquisition completion year

Variable Name	Definition
Model [3]:	
<i>CEO_TURN</i>	Indicator variable equal to one if the CEO was forced out within the three years following an M&A transaction; zero otherwise. Refer to the criteria following prior literature for classification of forced turnover within the methods section of the paper
<i>ΔOCF_NEGATIVE</i>	Indicator variable equal to one if the change in operating cash flows over the period from $t - 1$ to $t + 3$ is negative; zero otherwise
<i>ΔROA_{t-1}</i>	The change in net income scaled by average total assets of the acquiring company from year $t - 2$ to $t - 1$
<i>ΔSALES_{t-1}</i>	The change in sales scaled by average total assets of the acquiring company from year $t - 2$ to $t - 1$
<i>ΔRET_{t-1}</i>	The percentage change in the market value of equity of the acquiring company in year $t - 1$
<i>LOSS</i>	Indicator variable equal to one if net income is negative for the acquiring company in year $t - 1$; zero otherwise
<i>CEO_TENURE</i>	The number of years the CEO has held the position of CEO
<i>CEO_CHAIR</i>	Indicator variable equal to one if the CEO is also the Chairman of the Board of Directors; zero otherwise
Falsification and Robustness Tests:	
<i>UNFORCED</i>	Indicator variable equal to one if the CEO left the company within the three years following an M&A transaction and did not meet the criteria for classification as a forced turnover; zero otherwise. Refer to the criteria following prior literature for classification of forced turnover within the methods section of the paper
<i>ΔOCF_NEGATIVE</i>	Indicator variable equal to one if the change in operating cash flows over the period from $t - 1$ to $t + 3$ is negative; zero otherwise

Appendix 3. Motivation for H2b with a numerical illustration

By definition, the amount of goodwill recognized in a business combination is the difference between the fair value of the purchase consideration and the acquired net identifiable assets. The underlying intuition for H2b relies on the assumption that goodwill can represent either:

- Expected future benefits to the acquirer implied by the acquisition price from unidentifiable assets (e.g., the value of the target's human capital, competitive advantages in the marketplace, synergies arising from the business combination), or
- Overpayment (i.e., the acquirer paid more for the target than the expected benefits arising from the net identifiable assets).

To the extent that goodwill captures expected future benefits, the amount of goodwill should be positively correlated with the acquiring firm's future operating performance. On the other hand, the correlation between goodwill and future operating

performance will become less positive if the acquirer overpaid. In our tests of H2b, we expect that the likelihood of overpayment is higher among acquirers with M&A KAMs, which will manifest as a negative β_3 coefficient. We illustrate these effects using the following numerical example of two hypothetical acquisitions.

	(1) Acquirer A	(2) Acquirer B
Target identifiable net assets	15.00	15.00
Goodwill	5.00	5.00
Purchase consideration	20.00	20.00
Annual realized cash flow from target	2.00	0.75
Acquirer book value total assets	100.00	100.00
Δ cash flow/acquirer assets	2.0%	0.75%

Column (1) represents a cash acquisition made by Acquirer A of a target company for purchase consideration of \$20.0. The target's identifiable net assets have an estimated fair value of \$15.0, resulting in \$5.0 in goodwill. Assuming the target generates annual realized post-acquisition cash flows of \$2.0 and that the acquirer has total assets of \$100.0, this translates to an incremental cash-return-on-assets of 2.0%. Assuming a 5% discount rate in perpetuity, the expected benefits generated by the acquisition ($\$40.0 = \$2.0/5\%$) exceed the purchase price.

Column (2) represents a similar acquisition made by Acquirer B for purchase consideration of \$20.0 of a target that has identifiable net assets with a fair value of \$15.0, resulting in \$5.0 in goodwill. The difference is that the target in column (2) generates annual realized post-acquisition cash flows of \$0.75. Assuming that Acquirer B also has total assets with a book value of \$100.0, this translates to an incremental cash-return-on-assets of 0.75%. In this case, the expected benefits from the acquisition are \$15.0 assuming a perpetuity at the same 5% discount rate ($\$15.0 = \$0.75/5\%$). Therefore, Acquirer B overpaid for the target which results in a lower cash-return-on-assets compared to that for Acquirer A.

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Data Availability All data are available from publicly available sources.

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